Cochrane Library

Cochrane Database of Systematic Reviews

Cannabis-based medicines for chronic neuropathic pain in adults (Review)

Ateş G, Welsch P, Klose P, Phillips T, Lambers B, Häuser W, Radbruch L

Ateş G, Welsch P, Klose P, Phillips T, Lambers B, Häuser W, Radbruch L.
Cannabis-based medicines for chronic neuropathic pain in adults.
Cochrane Database of Systematic Reviews 2026, Issue 1. Art. No.: CD012182.
DOI: 10.1002/14651858.CD012182.pub3.

www.cochranelibrary.com

WILEY

TABLE OF CONTENTS

ABSTRACT 1
PLAIN LANGUAGE SUMMARY 3
SUMMARY OF FINDINGS 5
BACKGROUND 11
OBJECTIVES 13
METHODS 13
RESULTS 18
Figure 1. 19
Figure 2. 21
Figure 3. 22
DISCUSSION 27
AUTHORS' CONCLUSIONS 29
SUPPLEMENTARY MATERIALS 30
ADDITIONAL INFORMATION 30
REFERENCES 33
ADDITIONAL TABLES 41

[Intervention Review]

Cannabis-based medicines for chronic neuropathic pain in adults

Gülay Ateş[1], Patrick Welsch[2], Petra Klose[3], Tudor Phillips[4], Britta Lambers[5,6], Winfried Häuser[2,7], Lukas Radbruch[8]

[1]Institute for Digitalization and General Medicine, Center for Rare Diseases Aachen (ZSEA), Medical Faculty RWTH Aachen University, Aachen, Germany. [2]Medical Center Pain Medicine and Mental Health, Saarbrücken, Germany. [3]Internal and Integrative Medicine, Faculty of Medicine, Evang. Kliniken Essen-Mitte, University of Duisburg-Essen, Essen, Germany. [4]Pain Research and Nuffield Department of Clinical Neurosciences (Nuffield Division of Anaesthetics), University of Oxford, Pain Research Unit, Churchill Hospital, Oxford, UK. [5]UVSD SchmerzLOS e.V. - Independent Association of Pain Patients, Neumünster, Germany. [6]Departement of Health & Social, Hochschule Fresenius gemeinnützige Trägergesellschaft mbH - University of Applied Science, Cologne, Germany. [7]Department of Psychosomatic Medicine and Psychotherapy, Technische Universität München, München, Germany. [8]Palliative Medicine, University Hospital Bonn, Bonn, Germany

Contact: Winfried Häuser, winfriedhaeuser@googlemail.com.

Editorial group: Cochrane Central Editorial Service.
Publication status and date: Edited (no change to conclusions), published in Issue 1, 2026.

Citation: Ateş G, Welsch P, Klose P, Phillips T, Lambers B, Häuser W, Radbruch L. Cannabis-based medicines for chronic neuropathic pain in adults. *Cochrane Database of Systematic Reviews* 2026, Issue 1. Art. No.: CD012182. DOI: 10.1002/14651858.CD012182.pub3.

ABSTRACT

Rationale

Estimates of the population prevalence of chronic pain with neuropathic components range from 6% to 10%. Current pharmacological treatments for neuropathic pain help only a minority. New treatments are needed. Cannabis is increasingly promoted in the media as a treatment for chronic pain. This is an update of a review first published in 2018.

Objectives

To assess the benefits and harms of cannabis-based medicines (herbal, plant-based, synthetic) compared to placebo or conventional drugs for chronic neuropathic pain conditions in adults.

Search methods

We searched CENTRAL, MEDLINE, Embase, and three trial registries, together with reference checking. The latest search date was 29 January 2025.

Eligibility criteria

We selected randomised, double-blind controlled trials of medical cannabis, plant-derived and synthetic cannabinoids, against placebo or any other active treatment for chronic neuropathic pain conditions in adults, with a treatment duration of at least two weeks. We excluded studies whose double-blind duration was less than two weeks and studies which did not explicitly state that the pain was of a neuropathic nature.

Outcomes

Critical outcomes were the number of participants reporting pain relief of at least 50%, a Patient Global Impression of Change (PGIC) rating of 'much' or 'very much' improved, serious adverse events, and withdrawals due to adverse events.

Risk of bias

We assessed the risk of bias (RoB) for seven outcomes reported in three summary of findings tables using the Cochrane RoB 1 tool.

Synthesis methods

We synthesised results for each outcome using meta-analysis with a random-effects model by calculating absolute risk differences (RD) and standardised mean differences (SMD) with 95% confidence intervals (CI) for dichotomous outcomes and continuous outcomes, respectively. We used GRADE to assess the certainty of evidence for prespecified outcomes.

Included studies

We included six new studies involving 450 participants, along with 15 studies involving 1737 participants from the 2018 review, for a total of 21 studies with 2187 participants.

The studies ranged from two to 26 weeks in duration. Sample sizes ranged from 18 to 339 participants. Participants' mean age ranged from 34 to 61 years, and the proportion of women ranged from 0% to 90%.

Five studies included participants with central neuropathic pain, 14 studies included participants with peripheral neuropathic pain, and two studies included both types.

Seven studies administered tetrahydrocannabinol (THC)-dominant medicines; nine studies, balanced THC and cannabidiol (CBD) medicines; and five studies, CBD-dominant medicines. Twenty studies compared cannabis-based medicine to placebo, and one study's comparator was dihydrocodeine.

We judged the overall risk of bias to be low in six studies, unclear in 10 studies, and high in five studies.

Synthesis of results

THC-dominant medicines versus placebo. There is no clear evidence for an effect on pain relief of at least 50% (RD 0.14, 95% CI -0.07 to 0.37; 7 studies, 534 participants), PGIC rating of 'much' or 'very much' improved (RD 0.17, 95% CI -0.24 to 0.58; 2 studies, 72 participants), withdrawals due to adverse events (RD 0.03, 95% CI -0.02 to 0.08; 6 studies, 511 participants), serious adverse events (RD 0.02, 95% CI -0.01 to 0.06; 7 studies, 537 participants), pain relief of at least 30% (RD 0.16, 95% CI -0.08 to 0.40; 7 studies, 566 participants), and psychiatric disorder-related adverse events (RD 0.01, 95% CI -0.01 to 0.03; 4 studies, 368 participants), all with very low-certainty evidence. They may increase nervous system adverse events (RD 0.25, 95% CI 0.14 to 0.37; 5 studies, 439 participants; low-certainty evidence).

THC/CBD-balanced medicines versus placebo. There is no clear evidence for an effect on pain relief of at least 50% (RD 0.04, 95% CI 0.00 to 0.08; 8 studies, 746 participants) and serious adverse events (RD 0.01, 95% CI -0.02 to 0.03; 11 studies, 1449 participants), both with very low-certainty evidence. The evidence is very uncertain about the effect on nervous system-related (RD 0.39, 95% CI 0.23 to 0.55; 11 studies, 1445 participants) and psychiatric disorder-related adverse events (RD 0.08, 95% CI 0.03 to 0.13; 9 studies, 1375 participants), both very low-certainty evidence. They may increase PGIC ratings of 'much' or 'very much' improved (RD 0.07, 95% CI 0.02 to 0.11; 7 studies, 1145 participants), pain relief of at least 30% (RD 0.07, 95% CI 0.02 to 0.12; 10 studies, 1285 participants), and withdrawals due to adverse events (RD 0.05, 95% CI 0.02 to 0.09; 11 studies, 1449 participants), all with low-certainty evidence, though these effects were not clinically relevant.

CBD-dominant medicines versus placebo. There is no clear evidence for an effect on pain relief of at least 50% (RD -0.08, 95% CI -0.20 to 0.05; 5 studies, 208 participants; very low-certainty evidence). They may increase or decrease PGIC ratings of 'much' or 'very much' improved (RD -0.03, 95% CI -0.22 to 0.16; 2 studies, 79 participants), withdrawals due to adverse events (RD 0.02, 95% CI -0.03 to 0.06; 5 studies, 213 participants), serious adverse events (RD 0.02, 95% CI -0.03 to 0.06; 5 studies, 213 participants), pain relief of at least 30% (RD -0.04, 95% CI -0.17 to 0.09; 5 studies, 218 participants), nervous system-related adverse events (RD -0.03, 95% CI -0.10 to 0.03; 5 studies, 208 participants), and psychiatric disorder-related adverse events (RD -0.01, 95% CI -0.06 to 0.04; 5 studies, 208 participants), all with low-certainty evidence.

Authors' conclusions

There is no clear evidence for an effect of THC-dominant medicines on pain relief of 50% or greater, PGIC ratings of 'much' or 'very much' improved, withdrawals due to adverse events, and serious adverse events (very low-certainty evidence).

There is no clear evidence for an effect of THC/CBD-balanced medicines on pain relief of 50% or greater and serious adverse events (very low-certainty evidence). They may increase PGIC ratings of 'much' or 'very much' improved, and withdrawals due to adverse events (low-certainty evidence).

There is no clear evidence for an effect of CBD-dominant medicines on pain relief of 50% or greater (very low-certainty evidence). They may increase or decrease PGIC ratings of 'much' or 'very much' improved, serious adverse events, and withdrawals due to adverse events (low-certainty evidence).

Funding

No funding.

Registration

DOI 2018 review: 10.1002/14651858.CD012182.pub2

PLAIN LANGUAGE SUMMARY

What are the benefits and risks of cannabis-based medicines for adults with chronic neuropathic pain?

Key messages

- Due to a lack of robust evidence, the benefits and harms of cannabis-based medicines are unclear.
- Larger, well-designed studies including people with serious internal diseases (e.g. heart, kidney) and mental disorders are needed to give better estimates of the benefits and harms of cannabis-based medicines.

Introduction to the review topic

What is neuropathic pain?

Neuropathic pain is nerve pain coming from malfunctions of or damage to the nervous system (brain, spinal cord, peripheral nerves). It is different from pain messages that are carried along healthy nerves from damaged tissue (for example, a fall, cut, or arthritic knee).

How is the condition treated?

Neuropathic pain is treated with different types of medicines. Only a minority of people with neuropathic pain experience sufficient pain relief with the medications available.

Several products based on the cannabis plant have been suggested as treatments for neuropathic pain. These products include inhaled herbal cannabis and various medications (sprays, tablets, creams, transdermal patch (a medicated patch stuck to the skin)) containing active cannabis ingredients obtained from the plant, or made synthetically.

Cannabis-based products may contain: mainly tetrahydrocannabinol (THC), one main component of the cannabis flower; mainly cannabidiol (CBD), another main component of the cannabis flower; or a balanced ratio of THC and CBD.

Some people with neuropathic pain claim that cannabis-based products are effective for them. These claims are often highlighted in the media.

What did we want to find out?

We wanted to find out:

- which components of the cannabis flower (if any) help reduce neuropathic pain;
- if cannabis-based medicines cause any unwanted or harmful effects.

What did we do?

We searched for studies that compared different types of cannabis-based medicines (products including mainly THC or CBD and products including a balanced ratio of THC and CBD) against an inactive, fake medication, known as a 'placebo'.

We compared and summarised their results, and rated our confidence in the evidence, based on factors such as study methods and sizes. We analysed three comparisons:

- THC-dominant medicines versus placebo;
- THC/CBD-balanced medicines versus placebo;
- CBD-dominant medicines versus placebo.

What did we find?

We found 21 studies that involved 2187 people with different types of neuropathic pain. The studies were conducted in Asia, Europe, and North America. The smallest study had 18 participants; the largest, 339 participants. Participants' average age ranged from 34 to 70 years. The proportion of women ranged from 0% to 90%. Studies lasted between two and 26 weeks. Pharmaceutical companies funded 14 (or two-thirds) of the studies.

Main results

THC-dominant medicines versus placebo

It is unclear if medicines with mainly THC make any difference to the number of people who:

- experience pain relief of at least 30% or at least 50%;
- rate their condition to be much or very much improved;
- stop the medication due to unwanted effects;
- experience serious unwanted or harmful effects;
- experience unwanted psychological effects (e.g. confusion).

THC-dominant medicines may slightly increase the number of unwanted effects of the nervous system (e.g. dizziness) compared to placebo.

THC/CBD-balanced medicines versus placebo

It is unclear if medicines with a balanced THC/CBD ratio make any difference to the number of people who experience:

- pain relief of at least 50%;
- serious unwanted or harmful effects;
- unwanted effects of the nervous system (e.g. dizziness); and
- unwanted psychological effects (e.g. confusion).

They may increase the number of people who rate their condition to be much or very much improved, experience pain relief of at least 30%, and stop the medication due to unwanted effects.

CBD-dominant medicines versus placebo

It is unclear if medicines with mainly CBD make any difference to the number of people who experience pain relief of at least 50%.

We do not know whether CBD-dominant medicines increase or decrease the number of people who:

- rate their condition to be much or very much improved;
- stop the medication due to unwanted effects;
- experience serious unwanted or harmful effects;
- experience pain relief of at least 30%;
- experience unwanted psychological or nervous system-related effects.

This is because the evidence is limited, and the results vary between studies.

What are the limitations of the evidence?

Our confidence in the evidence is low to very low, and the results of further research could differ from the results of this review. Four main factors reduced our confidence in the evidence. Some studies did not clearly report how they were conducted. The results for many outcomes differed between studies. The results were also often not precise enough to draw firm conclusions. People with serious internal diseases (e.g. of the heart or liver) were excluded from the studies.

How current is the evidence?

The evidence is current to 29 January 2025.

SUMMARY OF FINDINGS

Summary of findings 1. THC-dominant medicines compared with placebo for chronic neuropathic pain

THC-dominant cannabis-based medicines compared with placebo for chronic neuropathic pain

Patient or population: adults with chronic neuropathic pain

Settings: outpatient study centres and hospitals in Europe and North America

Intervention: THC-dominant medicines (smoked THC cannabis flowers; oral plant-based THC (dronabinol) or synthetic tetrahydrocannabinol (THC) (nabilone)

Comparison: placebo

Outcomes	**Anticipated absolute effects (95% CI)**		Absolute risk difference (95% CI)	No. of participants (studies)	Certainty of evidence (GRADE)	Comments
	Risk with THC-dominant medicines	Risk with placebo				
Participant-reported pain relief of 50% or greater Assessed at end of study period (range: at day 10 to day 70) Higher proportion preferable	373 per 1000 (163 to 603)	233 per 1000	0.14 (-0.07 to 0.37)	534 (7 studies)	⊖⊖⊖⊖ **Very low**[a-d]	There is no clear evidence for an effect.
Patient Global Impression of Change rating of 'much' or 'very much' improved Assessed at end of study period (range: at day 10 to day 70) Higher proportion preferable	355 per 1000 (0 to 765)	185 per 1000	0.17 (-0.24 to 0.58)	72 (2 studies)	⊖⊖⊖⊖ **Very low**[a-d]	There is no clear evidence for an effect.
Withdrawals due to adverse events During entire study period Lower proportion preferable	57 per 1000 (7 to 107)	27 per 1000	0.03 (-0.02 to 0.08)	511 (6 studies)	⊕⊖⊖⊖ **Very low**[a,b,d]	There is no clear evidence for an effect.
Serious adverse events During entire study period	63 per 1000 (33 to 103)	43 per 1000	0.02 (-0.01 to 0.06)	537 (7 studies)	⊕⊖⊖⊖ **Very low**[a,b,d]	There is no clear evidence for an effect.

Lower proportion preferable						
Participant-reported pain relief of 30% or greater Assessed at end of study period (range: day 10 to day 70) Higher proportion preferable	565 per 1000 (325 to 805)	405 per 1000	0.16 (-0.08 to 0.40)	566 (7 studies)	⊝⊝⊝⊝ **Very low**[a-d]	There is no clear evidence for an effect.
Specific adverse events: nervous system disorder During entire study period Lower proportion preferable	476 per 1000 (366 to 596)	226 per 1000	0.25 (0.14 to 0.37)	439 (5 studies)	⊕⊕⊝⊝ **Low** [a,d]	NNTH 4 (95% CI 3 to 6) The effect is clinically relevant. THC-dominant medicines may increase adverse events related to the nervous system.
Specific adverse events: psychiatric disorders During entire study period Lower proportion preferable	29 per 1000 (9 to 49)	19 per 1000	0.01 (-0.01 to 0.03)	368 (4 studies)	⊕⊝⊝⊝ **Very low**[a,b,d]	There is no clear evidence for an effect.

CI: confidence interval; **NNTB**: number needed to treat for an additional beneficial outcome; **NNTH**: number needed to treat for an additional harmful outcome; **THC:** tetrahydrocannabinol

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate; the true effect is likely to be close to the estimate of effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited; the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate; the true effect is likely to be substantially different from the estimate of effect.

[a]Downgraded by one level for indirectness: most studies excluded people with current or historical mental disorders and major medical diseases.
[b]Downgraded by one level for imprecision: 95% CI is wide, includes the possibility of no effect, or both.
[c]Downgraded by one level for inconsistency: I^2 exceeded 40%.
[d]Downgraded by one level for risk of bias: most studies were at an overall high or unclear risk of bias.

Summary of findings 2. THC/CBD-balanced medicines compared with placebo for chronic neuropathic pain

Patient or population: adults with chronic neuropathic pain

Settings: outpatient study centres and hospitals in Europe and North America

Intervention: THC/CBD-balanced medicines (oral or oromucosal plant-derived THC and CBD

Comparison: placebo

Outcomes	Anticipated absolute effects (95% CI)		Absolute risk difference (95% CI)	No. of participants (studies)	Certainty of evidence (GRADE)	Comments
	Risk with THC/CBD- balanced medicines	Risk with placebo				
Participant-reported pain relief of 50% or greater Assessed at end of study period (range: day 10 to day 105) Higher proportion preferable	239 per 1000 (177 to 257)	177 per 1000	0.04 (0.00 to 0.08)	746 (8 studies)	⊕⊝⊝⊝ **Very low**[a,b,d]	There is no clear evidence for an effect.
Patient Global Impression of Change rating of 'much' or 'very much' improved Assessed at end of study period (range: day 10 to day 105) Higher proportion preferable	268 per 1000 (218 to 308)	197 per 1000	0.07 (0.02 to 0.11)	1145 (7 studies)	⊕⊕⊝⊝ **Low**[a,b]	NNTB 16 (95% CI 9 to 50) The effect is clinically not relevant. THC/CBD-balanced medicines may slightly increase the number of participants with a PGIC rating of 'much' or 'very much' improved.
Withdrawals due to adverse events During entire study period Lower proportion preferable	118 per 1000 (88 to 158)	68 per 1000	0.05 (0.02 to 0.09)	1449 (11 studies)	⊕⊕⊝⊝ **Low**[a,b]	NNTH 20 (95% CI 11 to 50) The effect is clinically not relevant. THC/CBD-balanced medicines may slightly increase the number of participants withdrawing from the study due to adverse events.
Serious adverse events During entire study period	65 per 1000 (35 to 85)	55 per 1000	0.01 (-0.02 to 0.03)	1449 (11 studies)	⊕⊝⊝⊝ **Very low**[a,b,d]	There is no clear evidence for an effect.

Lower proportion preferable						
Participant-reported pain relief of 30% or greater Assessed at end of study period (range: day 10 to day 105) Higher proportion preferable	406 per 1000 (356 to 456)	336 per 1000	0.07 (0.02 to 0.12)	1285 (10 studies)	⊕⊕⊝⊝ **Low**[a,d]	NNTB 16 (95% CI 8 to 50) The effect is clinically not relevant. THC/CBD-balanced medicines may increase the number of participants with pain relief of 30% or greater.
Specific adverse events: nervous system disorder During entire study period Lower proportion preferable	644 per 1000 (484 to 804)	254 per 1000	0.39 (0.23 to 0.55)	1445 (11 studies)	⊕⊝⊝⊝ **Very low**[a,c,d]	NNTH 3 (95% CI 2 to 4) The effect is clinically relevant. The evidence is very uncertain about the effect.
Specific adverse events: psychiatric disorders During entire study period Lower proportion preferable	107per 1000 (57 to 157)	27 per 1000	0.08 (0.03 to 0.13)	1375 (9 studies)	⊕⊝⊝⊝ **Very low**[a,c,d]	NNTH 13 (95% CI 6 to 33) The effect is clinically not relevant. The evidence is very uncertain about the effect.

CBD: cannabidiol; **CI**: confidence interval; **NNTB**: number needed to treat for an additional beneficial outcome; **NNTH**: number needed to treat for an additional harmful outcome; **THC:** tetrahydrocannabinol

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate; the true effect is likely to be close to the estimate of effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited; the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate; the true effect is likely to be substantially different from the estimate of effect.

[a]Downgraded by one level for indirectness: most studies excluded people with current or historical mental disorders and major medical diseases.
[b]Downgraded by one level for imprecision: 95% CI is wide, includes the possibility of no effect, or both.
[c]Downgraded by one level for inconsistency: I^2 exceeded 40%.
[d]Downgraded by one level for risk of bias: most studies were at an overall high or unclear risk of bias.

Summary of findings 3. CBD-dominant medicines compared with placebo for chronic neuropathic pain

Patient or population: adults with chronic neuropathic pain

Settings: outpatient study centres and hospitals in Europe and North America

Intervention: CBD-dominant medicines (oral, cream)

Comparison: placebo

Outcomes	Anticipated absolute effects		Absolute risk difference (95% CI)	No. of participants (studies)	Certainty of evidence (GRADE)	Comments
	Risk with CBD-dominant medicines (95% CI)	Risk with placebo				
Participant-reported pain relief of 50% or greater Assessed at end of study period (range: day 10 to day 70) Higher proportion preferable	129 per 1000 (9 to 259)	209 per 1000	-0.08 (-0.20 to 0.05)	208 (5 studies)	⊕⊝⊝⊝ **Very low**[a,b,c]	There is no clear evidence for an effect.
Patient Global Impression of Change rating of 'much' or 'very much' improved Assessed at end of study period (range: day 10 to day 70) Higher proportion preferable	170 per 1000 (0 to 360)	200 per 1000	-0.03 (-0.22 to 0.16)	79 (2 studies)	⊕⊕⊝⊝ **Low**[a,b]	CBD-dominant medicines may increase or decrease the number of participants with a PGIC rating of 'much' or 'very much' improved.
Withdrawals due to adverse events During entire study period Lower proportion preferable	20 per 1000 (0 to 60)	0 per 1000	0.02 (-0.03 to 0.06)	213 (5 studies)	⊕⊕⊝⊝ **Low**[a,b]	CBD-dominant medicines may increase or decrease the number of participants withdrawing due to adverse events.
Serious adverse events During entire study period Lower proportion preferable	20 per 1000 (0 to 60)	0 per 1000	0.02 (-0.03 to 0.06)	213 (5 studies)	⊕⊕⊝⊝ **Low**[a,b]	CBD-dominant medicines may increase or decrease the number of participants with serious adverse events.
Participant-reported pain relief of 30% or greater Assessed at end of study period (range: day 10 to day 70)	274 per 1000 (144 to 404)	314 per 1000	-0.04 (-0.17 to 0.09)	208 (5 studies)	⊕⊕⊝⊝ **Low**[a,b]	CBD-dominant medicines may increase or decrease the number of participants with pain relief of 30% or greater.

Higher proportion preferable						
Specific adverse events: nervous system disorder During entire study period Lower proportion preferable	135 per 1000 (65 to 195)	165 per 1000	-0.03 (-0.10 to 0.03)	208 (5 studies)	⊕⊕⊝⊝ **Low**[a,b]	CBD-dominant medicines may increase or decrease the number of participants with adverse events of the nervous system.
Specific adverse events: psychiatric disorders During entire study period Lower proportion preferable	25 per 1000 (0 to 75)	35 per 1000	-0.01 (-0.06 to 0.04)	208 (5 studies)	⊕⊕⊝⊝ **Low**[a,b]	CBD-dominant medicines may increase or decrease the number of participants with adverse events related to psychiatric disorders.

CBD: cannabidiol; **CI**: confidence interval; **NNTB**: number needed to treat for an additional beneficial outcome; **NNTH**: number needed to treat for an additional harmful outcome

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate; the true effect is likely to be close to the estimate of effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited; the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate; the true effect is likely to be substantially different from the estimate of effect.

[a]Downgraded by one level for indirectness: most studies excluded people with current or historical mental disorders and major medical diseases.
[b]Downgraded by one level for imprecision: 95% CI is wide, includes the possibility of no effect, or both.
[c]Downgraded by one level for inconsistency: I^2 exceeded 40%.

BACKGROUND

The protocol for this review was based on a template for reviews of medication used to relieve neuropathic pain of the Cochrane Pain, Palliative and Supportive Care Group (which closed in 2023). The aim is for all reviews to use the same methods, based on established criteria for what constitutes reliable evidence in chronic pain [1]. This is the first update of a Cochrane review first published in 2018 [2]. The update was necessary because new studies have been published since 2018.

Description of the condition

The International Association for the Study of Pain defines neuropathic pain as "pain caused by a lesion or disease of the somatosensory system" [3]. Neuropathic pain is a consequence of a pathological maladaptive response of the nervous system to 'damage' from a wide variety of potential causes. It is characterised by pain in the absence of a noxious stimulus. It may be spontaneous (continuous or paroxysmal) in its temporal characteristics or be evoked by sensory stimuli (dynamic mechanical allodynia where pain is evoked by light touch of the skin). Neuropathic pain may involve sensory loss (such as numbness) or sensory gain (such as allodynia), with patterns that vary widely across individuals and conditions. These differences may reflect distinct underlying pain mechanisms in each person and may help predict which treatments will be most effective [4, 5, 6]. Pre-clinical research offers a bewildering array of possible pain mechanisms that may operate in people with neuropathic pain, which largely reflect pathophysiological responses in both the central and peripheral nervous systems, including neuronal interactions with immune cells [6, 7, 8].

The diagnosis of chronic neuropathic pain requires a history of nervous system injury or disease and a neuroanatomically plausible distribution of the pain. Negative (e.g. decreased or loss of sensation) and positive sensory symptoms or signs (e.g. allodynia or hyperalgesia) indicating the involvement of the somatosensory nervous system must be compatible with the innervation territory of the affected nervous structure [9].

The new classification of chronic neuropathic pain in the *International Classification of Diseases* Version 11 (ICD-11) lists the most common conditions of peripheral neuropathic pain: trigeminal neuralgia, peripheral nerve injury, painful polyneuropathy, postherpetic neuralgia, and painful radiculopathy. Conditions of central neuropathic pain include pain caused by spinal cord or brain injury, post-stroke pain, and pain associated with multiple sclerosis [9].

In systematic reviews, the overall prevalence of neuropathic pain in the general population is reported to be between 7% and 10% [10], and about 7% in a systematic review of studies published since 2000 [11]. In individual countries, prevalence rates have been reported as 3.3% in Austria [12], 6.9% in France [13], and up to 8% in the UK [14]. Some forms of neuropathic pain, such as painful diabetic polyneuropathy and post-surgical chronic pain, are increasing [15].

Estimates of incidence vary between individual studies for particular origins of neuropathic pain, often because of small numbers of cases. In primary care in the UK, between 2002 and 2005, the incidences (per 100,000 person-years' observation) were 28 (95% confidence interval (CI) 27 to 30) for postherpetic neuralgia, 27 (95% CI 26 to 29) for trigeminal neuralgia, 0.8 (95% CI 0.6 to 1.1) for phantom limb pain, and 21 (95% CI 20 to 22) for painful diabetic polyneuropathy [15]. Other studies have estimated an incidence of four in 100,000 per year for trigeminal neuralgia [16, 17], 12.6 per 100,000 person-years for trigeminal neuralgia, and 3.9 per 100,000 person-years for postherpetic neuralgia in a study of facial pain in the Netherlands [18]. One systematic review of chronic pain demonstrated that some neuropathic pain conditions, such as painful diabetic polyneuropathy, can be more common than other neuropathic pain conditions, with prevalence rates of up to 400 per 100,000 person-years [19].

Many people with neuropathic pain conditions are significantly disabled with moderate or severe pain for many years. Chronic pain caused by neuropathic conditions is responsible for considerable loss of quality of life and employment, and increased healthcare costs [11]. A study in the USA found that healthcare costs were three-fold higher for people with neuropathic pain than matched control participants [20]. A UK study and a German study reported that people with neuropathic pain used healthcare services at levels twofold to threefold higher than those without [21, 22]. For postherpetic neuralgia, for example, studies demonstrate a large loss of quality of life and substantial costs [23].

Neuropathic pain is difficult to treat effectively, with only a minority of people experiencing a clinically relevant benefit from any one intervention [1, 24]. A multidisciplinary approach is now advocated, combining pharmacological with physical or cognitive interventions, or both. Conventional analgesics such as paracetamol and nonsteroidal anti-inflammatory drugs (NSAIDs) are not thought to be effective, but without evidence to support or refute that view [25]. Some people may derive some benefit from a topical lidocaine patch or low-concentration topical capsaicin, although evidence about benefits is uncertain [26, 27]. High-concentration topical capsaicin may benefit some people with postherpetic neuralgia [28]. Treatment is often by so-called pain modulators, such as antidepressants (duloxetine and amitriptyline [29, 30, 31, 32]) or antiepileptics (gabapentin or pregabalin [33, 34, 35, 36]). Evidence for the efficacy of opioids is unconvincing [37, 38, 39].

The proportion of people who achieve worthwhile pain relief (typically at least 50% pain intensity reduction [40]) is small, generally only 10% to 25% more than with placebo, with numbers needed to treat for an additional beneficial outcome (NNTB) usually between four and 10 [1, 24]. Neuropathic pain is not particularly different from other chronic pain conditions in that only a small proportion of trial participants have a good response to treatment [1].

There is a need to explore other treatment options, with different mechanisms of action and from different drug categories, for treatment of neuropathic pain syndromes. Some patient organisations and advocates for the treatment of chronic pain refractory to conventional treatment have promoted medical cannabis, which is available for pain management in a growing number of countries in the world [41]. However, the use of cannabis for medical reasons is highly contested because of the adverse health effects of long-term cannabis use for recreational purposes [42].

Description of the intervention and how it might work

The endocannabinoid system is ubiquitous in the animal kingdom and is said to perform multiple functions that move the organism back to equilibrium. A large body of evidence currently supports the presence of cannabinoid (CB) receptors and ligands in the peripheral and central nervous system, but also in other tissues such as bone and in the immune system [43].

The endocannabinoid system has three broad and overlapping functions in mammals. The first is a stress recovery role, operating in a feedback loop in which endocannabinoid signalling is activated by stress and functions to return endocrine, nervous, and behavioural systems to homeostatic balance. The second is to control energy balance through regulation of the intake, storage, and utilisation of food. The third involves immune regulation: endocannabinoid signalling is activated by tissue injury and modulates immune and inflammatory responses [44]. Thus, the endocannabinoid neuromodulatory system appears to be involved in multiple physiological functions, such as anti-nociception, cognition and memory, endocrine function, nausea and vomiting, inflammation, and immune recognition [44].

Cannabis is a genus of flowering plants in the family Cannabaceae. The number of species within the genus is disputed. Typically, three main species are recognised: *Cannabis sativa*, *Cannabis indica,* and *Cannabis ruderalis*. These plants, commonly known as marijuana, have been used for pain relief for millennia, and have additional effects on appetite, sleep, and mood [45]. Cannabinoids are C21 terpenophenolic compounds found in *C sativa*, including structurally related analogs and transformation products. Chemically, these compounds combine terpene-like and phenol-like structures, and the term also encompasses variants formed through chemical or metabolic changes. More broadly, 'cannabinoids' refers to all ligands of cannabinoid receptors and their related compounds, including endogenous receptor ligands and synthetic analogs. Based on their source, cannabinoids are classified into three types: (a) endocannabinoids, endogenous derivatives of polyunsaturated fatty acids that act as natural signalling molecules in animals; (b) phytocannabinoids, derived from plants; and (c) synthetic cannabinoids, which are chemically produced analogs [46].

Cannabis contains over 450 compounds, with at least 120 classified as phytocannabinoids. Delta-9-tetrahydrocannabinol (THC) and cannabidiol (CBD) are the most well-studied phytocannabinoids and are of particular medical interest. THC is the main active constituent, with psychoactive (e.g. reduction of anxiety and stress) and pain-relieving properties. Tetrahydrocannabinol (Δ8-THC), and cannabinol (CBN) are examples of other psychoactive cannabinoids found in cannabis plants. The second molecule of interest is cannabidiol (CBD), which has lower affinity for the cannabinoid (CB) receptors and the potential to counteract the negative effects of THC on memory, mood, and cognition, but may have an effect on pain modulation by anti-inflammatory properties. Because the psychotropic effects of THC are a concern, there is growing interest in other phytocannabinoids that lack psychotropic activity, such as cannabichromene (CBC), cannabigerol (CBG), and cannabidivarin (CBDV). CBDV is a minor cannabinoid, being a propyl analogue of CBD. Like CBD and unlike THC, CBDV has very low affinity for CB_1 and CB_2 receptors. It acts as an agonist of Transient Receptor Potential Ankyrin 1 (TRPA1) and Transient Receptor Potential cation channel subfamily V members 1 and 2 (TRPV1 and TRPV2), as an inverse agonist of G protein-coupled receptor 6 (GPR6), and as an antagonist of G protein-coupled receptor 55 (GPR55) [47]. The specific roles of currently identified endocannabis-based medicines that act as ligands at CB receptors within the nervous system (primarily but not exclusively CB_1 receptors) and in the periphery (primarily but not exclusively CB_2 receptors) are only partially elucidated, but there are abundant pre-clinical data to support their influence on nociception [43].

It is also hypothesised that cannabis reduces alterations in cognitive and autonomic processing in chronic pain states [48]. The frontal-limbic distribution of CB receptors in the brain suggests that cannabis may preferentially target the affective qualities of pain [49]. In addition, cannabis may attenuate low-grade inflammation, another postulate for the pathogenesis of neuropathic pain [50].

Cannabis-based medicines

The terminology and definitions associated with cannabis-based medicines vary in the literature. In Supplementary material 7, we provide terminology based on the proposals by taskforces of the European Pain Federation and the International Association for the Study of Pain [51, 52]. In this review, we use the term 'cannabis-based medicines' to encompass registered, regulatory body-approved medicinal cannabis extracts with defined and standardised phytocannabinoid content or standardised synthetic cannabinoid content, as well as medical cannabis.

Cannabis-based medicines are available in different forms. Licenced drugs and products currently being tested for medical use include:

- plant-derived products with a defined dosage of plant-extracted phytocannabinoids such as CBD, THC, CBDV, or CBN, and their combinations;
- plant-derived products with herbal cannabis with a defined dosage of phytocannabinoids such as CBD, THC, CBDV, or CBN and their combinations, together with other less defined components of the cannabis plant (terpenes and flavonoids) – so-called 'full spectrum' products;
- synthetic cannabinoids such as the THC analogues nabilone and dronabinol.

Some cannabis-based medicines are available as a finished medical product; for example, an oromucosal spray containing defined dosages of THC and CBD per actuation (e.g. nabiximols). Other cannabis-based medicines are available as magistral preparations (any medicinal product prepared in a pharmacy in accordance with the medical prescription for an individual patient), such as so-called 'full spectrum' cannabis products extracted from the cannabis flower. The content of THC and CBD in medical cannabis is highly variable and ranges from 1% to 28% THC and 0% to 20% CBD.

The main routes of administration are topical (dermal, oromucosal, rectal), oral (pills, oil), and inhalation (smoking, vaporisation).

There are substantial differences between countries in the availability of cannabis-based medicines for medical use and in whether these medicines are covered by public health systems [41, 53]. In some countries, plant-derived products with defined concentrations of phytocannabinoids (e.g. CBD and less than 0.2% THC) are available as nutritional supplements.

Taking into consideration the poorly understood pathogenesis of chronic neuropathic pain syndromes, the complexity of symptom expression, and the absence of an ideal treatment, the potential for manipulation of the cannabinoid system as a therapeutic modality is attractive.

Why it is important to do this review

While experts generally agree on the role of antidepressants and anticonvulsants in the management of chronic neuropathic pain [36], the use of opioids and cannabis-based medicines remains the subject of debate [36, 54]. For example, systematic reviews of the use of cannabis-based medicines to treat chronic pain, published between 2010 and 2019, came to different conclusions about their importance in chronic neuropathic pain [55, 56]. This divergence was probably due to the inclusion of different trials, different standards to evaluate the quality of evidence, and different weighting of the outcomes of efficacy, tolerability, and safety. In a 2022 commentary in the *European Journal of Pain* that considered the state of the field, Eisenberg and colleagues described a 'Bermuda triangle' of low-quality studies in cannabis-based medicines for chronic pain, countless meta-analyses of those studies, leading to conflicting recommendations [57]. At the same time, professional pain organisations have made contrasting recommendations for the use of cannabis-based medicines for chronic pain. A presidential taskforce of the International Association for the Study of Pain on cannabis and cannabinoid analgesia concluded that "due to lack of high-quality clinical evidence, the use of cannabis/CBM is not currently endorsed for pain relief" [58]. However, a European Pain Federation taskforce stated: "Therapy with cannabis-based medicines should only be considered by experienced clinicians as part of a multidisciplinary treatment and preferably as adjunctive medication if guideline-recommended first- and second-line therapies have not provided sufficient efficacy or tolerability. The quantity and quality of evidence are such that cannabis-based medicines may be reasonably considered for chronic neuropathic pain. For all other chronic pain conditions (cancer, non-neuropathic noncancer pain), the use of cannabis-based medicines should be regarded as an individual therapeutic trial" [51].

Due to the conflicting conclusions of systematic reviews and the differing positions of international pain societies on the importance of cannabis-based medicines in treating chronic neuropathic pain [59, 60], as well as the public debate on the medical use of herbal cannabis for chronic pain [61], we saw the need for a Cochrane review applying the standards of the Cochrane Pain, Palliative and Supportive Care (PaPaS) Group.

OBJECTIVES

To assess the benefits and harms of cannabis-based medicines (herbal, plant-based, synthetic) compared to placebo or conventional drugs for chronic neuropathic pain conditions in adults.

METHODS

Differences between the first and second versions of the review are as follows.

- We adopted the new classification of chronic neuropathic pain syndromes developed by the International Association for the Study of Pain (IASP) for the *International Classification of Diseases* (ICD-11) [9].
- We assessed the trustworthiness of the studies included in the first version of the review and the newly included studies using the Research Integrity Assessment tool developed by Weibel and colleagues [62].
- We removed fatigue as an outcome because it was reported by very few studies.
- We applied stricter criteria for two risk of bias domains: we rated studies as having a low risk of selection bias only when they used block randomisation, and as having a low risk of reporting bias only when authors had published a protocol before the study began.
- We refined the criteria for the 'incomplete outcome data' risk of bias domain by incorporating attrition rates into our judgements, in line with established guidance [63].
- We defined effect sizes for dichotomous outcomes using risk differences, expressed as NNTB/NNTH (number needed to treat for an additional beneficial/harmful outcome), and interpreted them according to predefined thresholds for clinical relevance.
- We stratified the analyses by type of cannabis-based medicine (i.e. THC-dominant, THC/CBD-balanced, CBD-dominant).
- In the first version of the review [2], we performed subgroup analyses by type of cannabis-based medicines. In this update, we stratified all analyses primarily by cannabis type and omitted the remaining subgroup analyses to avoid inflating the total number of subgroup comparisons [64].
- We defined a sensitivity analysis by excluding studies flagged as ineligible or 'awaiting classification' by the Research Integrity Assessment tool [62]. Additionally, we conducted a post hoc sensitivity analysis excluding one outlier study and studies with double-blind durations of less than four weeks.

We followed the Methodological Expectations of Cochrane Intervention Reviews (MECIR) standards in conducting this review and reporting our findings [65, 66], as well as PRISMA 2020 for reporting the review [67].

Criteria for considering studies for this review

Types of studies

We included randomised controlled trials (RCTs) of at least two weeks' double-blind duration (drug titration and maintenance or withdrawal). We included studies with parallel-group, cross-over, or enriched-enrolment randomised withdrawal designs, considering only the double-blind phase results, with at least 10 participants per treatment arm. We analysed the data of the first treatment period of cross-over trials (if reported) to avoid carry-over effects. We generally required full journal publications, although we also included online clinical trial results summaries of unpublished trials and abstracts that reported sufficient data for analysis. We included studies that reported at least one benefit and one harm outcome (as defined below) amenable to quantitative analysis.

We excluded studies if they:

- were published only as short abstracts;
- were not randomised;
- involved experimental pain (i.e. where pain is induced under controlled laboratory conditions rather than arising naturally from a medical condition), case reports, or clinical observations;

- had less than two weeks of double-blind follow-up;
- enroled fewer than 10 participants per treatment arm.

We excluded studies in people with multiple sclerosis with spasticity as the primary outcome and pain as a secondary outcome if such studies did not report in their methods section that the pain assessed was of a neuropathic nature. In a cohort study of 302 people with multiple sclerosis, 30% had chronic pain. The pain was of a neuropathic nature in half of these participants [68]. A committee of the American Academy of Neurology stated that spasticity-associated pain should not be mixed with central neuropathic pain in multiple sclerosis [69]. We excluded studies in people with fibromyalgia because the nature of fibromyalgia (neuropathic or not) is under debate [70]. Additionally, cannabis-based medicines for fibromyalgia are the subject of another Cochrane review [71]. We excluded studies in people with 'mixed pain' because this review focuses only on neuropathic pain [72].

As noted above and in Selection of studies, we assessed all trials meeting the review's eligibility criteria using the Research Integrity Assessment tool.

Types of participants

Studies included adults aged 18 years and above with one or more chronic (three months or longer) neuropathic pain condition, including (but not limited to):

- cancer-related neuropathy;
- central neuropathic pain (e.g. multiple sclerosis);
- complex regional pain syndrome (CRPS) Type II;
- HIV neuropathy;
- painful diabetic neuropathy;
- peripheral polyneuropathy of other aetiologies; for example, toxic (alcohol, drugs);
- phantom limb pain;
- postherpetic neuralgia;
- postoperative or traumatic peripheral nerve lesions;
- spinal cord injury;
- nerve plexus injury;
- trigeminal neuralgia.

Where included studies had participants with more than one type of neuropathic pain, we analysed results according to the primary condition. Studies had to state explicitly that they included people with neuropathic pain (by title and in their methods section).

Types of interventions

Eligible interventions were cannabis-based medicines, including herbal cannabis (hashish, marijuana), plant-based cannabinoids (dronabinol; nabiximols), and pharmacological (synthetic) cannabinoids (e.g. nabilone), at any dose, by any route, administered for the relief of neuropathic pain and compared to placebo or any active comparator.

Following McDonagh and colleagues [73], we categorised cannabis-based medicines in this update by THC-to-CBD ratio:

- THC-dominant: THC/CBD ratio > 2:1
- THC/CBD-balanced: THC/CBD ratio < 2:1 and > 1:2
- CBD-dominant: CBD/THC ratio > 2:1.

For analysis, we combined cannabidiol (CBD) and cannabidivarin (CBDV).

In addition, we specified whether the medications were synthetic (e.g. nabilone) or derived from whole-plant cannabis. The term 'dronabinol' is used for synthetic THC analogues as well as for plant-extracted medications. Extracted products may contain other cannabinoids and other compounds that may or may not influence the effect of the intervention.

We did not include studies in the following experimental medications which have not yet been approved or are not available for use in pain management outside clinical studies, or both [74].

- Drugs that manipulate the endocannabinoid system by inhibiting enzymes that hydrolyse endocannabinoids and thereby boost the levels of the endogenous molecules (e.g. blockade of the catabolic enzyme fatty acid amide hydrolase (FAAH)) [75].
- Levonantradol, a synthetic THC analogue, which is not available for medical use in any country in the world.
- Cannabinoid (CB) receptor antagonists and negative allosteric modulators (e.g. rimonabant (SR141716A)).

We created these comparisons:

- Cannabis-based medicines, stratified by type of cannabis-based medicine, versus placebo in studies with parallel-group, cross-over, or enriched-enrolment randomised withdrawal designs, considering only the double-blind phase results.
- Cannabis-based medicines, stratified by type of cannabis-based medicine, versus active pharmacological intervention in studies with parallel-group, cross-over, or enriched-enrolment randomised withdrawal designs, considering only the double-blind phase results.

Outcome measures

The standards used to assess evidence in chronic pain trials have changed substantially in recent years, with particular attention being paid to trial duration, withdrawals, and statistical imputation following withdrawal, all of which can substantially alter estimates of efficacy. The most important change is the move from using mean pain scores, or mean change in pain scores, to the number of people who have a large decrease in pain (at least 50%) and who continue in treatment, ideally in trials of eight to 12 weeks' duration or longer. These standards are set out in the 'PaPaS Author and Referee Guidance' for pain studies of Cochrane Pain, Palliative and Supportive Care [76]. We assessed evidence using (a) methods that make both statistical and clinical sense and (b) criteria for what constitutes reliable evidence in chronic pain [1].

We were particularly interested in the Initiative on Methods, Measurement, and Pain Assessment in Clinical Trials (IMMPACT) definitions of moderate and substantial benefit in chronic pain studies [77], which provide consensus thresholds for clinically meaningful improvements in pain intensity, physical functioning, and patient-reported global change.

Critical outcomes

- Participant-reported pain relief of 50% or greater (at the end of the study period). We preferred composite neuropathic pain scores over single-scale generic pain scores if studies used both measures.

- Patient Global Impression of Change (PGIC) rating of 'much' or 'very much' improved (at the end of the study period).
- Withdrawals due to adverse events (tolerability) (during the entire study period).
- Serious adverse events (harm) (during the entire study period). We adapted the International Council for Harmonisation's definition of serious adverse events as typically including "any untoward medical occurrence that at any dose results in death, is life-threatening, requires inpatient hospitalisation or prolongation of existing hospitalisation, [or] results in persistent or significant disability/incapacity", as well as any "important medical event that may jeopardise the person, or may require an intervention to prevent one of the above characteristics/ consequences" [78].

Important outcomes

- Participant-reported pain relief of 30% or greater (at the end of the study period). We preferred composite neuropathic pain scores over single-scale generic pain scores if studies used both measures.
- Specific adverse events, particularly nervous system (e.g. dizziness, somnolence) and psychiatric disorders (e.g. confusion, paranoia, psychosis, substance dependence) according to the Medical Dictionary for Regulatory Activities (MedDRA) [79].
- Mean pain intensity (at the end of the study period). We preferred composite neuropathic pain scores over single-scale generic pain scores if studies used both measures.
- Health-related quality of life (at the end of the study period). We preferred composite scores (e.g. 36-item Short Form Health Survey (SF-36)) over single generic scores if studies used both measures.
- Sleep problems (at the end of the study period). We preferred composite scores (e.g. Leeds Sleep Evaluation Questionnaire) over single generic scores if studies used both measures.
- Psychological distress (at the end of the study period). We preferred composite scores (e.g. Hospital Anxiety and Depression Scale) over single generic scores if studies used both measures. If studies reported both depression and anxiety scores, we used depression scores for analysis.
- Withdrawals due to lack of efficacy (during the entire study period).
- Any adverse event (during the entire study period).

Search methods for identification of studies

Electronic searches

Our information specialist (PK) conducted systematic searches of the following sources from 7 November 2017 (date of the last search for the 2018 version of the review [2]) to 29 January 2025 (date of last search for all databases) with no restrictions on the language of publication.

- The Cochrane Central Register of Controlled Trials (CENTRAL) via the Cochrane Register of Studies Online 2025, Issue 1 [80]
- MEDLINE (via PubMed) (7 November 2017 to 29 January 2025)
- Scopus Elsevier (7 November 2017 to 29 January 2025)

Supplementary material 1 shows the search strategies.

Searching other resources

We reviewed the bibliographies of retrieved RCTs and review articles. We searched the following clinical trials databases to identify additional published or unpublished data:

- US National Institutes of Health clinical trial register (ClinicalTrials.gov);
- European Union Clinical Trials Register (www.clinicaltrialsregister.eu);
- World Health Organization (WHO) International Clinical Trials Registry Platform (ICTRP) (apps.who.int/trialsearch/).

We contacted trial investigators to request missing data. Our research integrity assessment included a search in the Retraction Watch Database of studies included in the analyses. We searched PubMed for post-publication amendments of studies included in the analyses.

Data collection and analysis

We performed separate analyses of the three types of cannabis-based medicines: THC-dominant, THC/CBD-balanced, and CBD-dominant.

Selection of studies

Two review authors (PW, WH) independently determined eligibility by reading the abstract of each study identified by the search. We eliminated studies that clearly did not satisfy the inclusion criteria, and obtained full copies of the remaining studies. Two review authors (PW, WH) independently read these studies and reached agreement about eligibility through discussion. We did not anonymise the studies before the assessment. We documented the trial selection process in a PRISMA flow diagram with the total number of trials included, excluded, awaiting classification, and ongoing [81]. We listed the reasons for exclusion and awaiting classification in the Characteristics of excluded studies (Supplementary material 3) and Characteristics of studies awaiting classification (Supplementary material 4) tables.

Research integrity screening

People with chronic pain, clinicians, and the public are sometimes poorly served by an evidence architecture that contains multiple structural weaknesses, such as misconduct or lack of research integrity [82]. Cochrane defines a 'problematic study' as any "published or unpublished study where there are serious questions about the trustworthiness of the data or findings, regardless of whether the study has been formally retracted; scientific misconduct will not be the only reason that a study might be problematic; problems may result from poor research practices or honest errors" [83]. The Research Integrity Assessment (RIA) tool developed by Weibel and colleagues evaluates research integrity using six domains: retraction notices, prospective trial registration, ethics approval, plausible authorship, sufficient methodological reporting of eligibility criteria (e.g. randomisation), and the plausibility of study results [62]. Weibel and colleagues recommend that failure in any of the first three domains – i.e. retraction, no prospective registration, no adequate ethics approval – should lead to exclusion of an RCT. For domains 4 through 6 (author group, methods, results), the authors recommend more caution: if concerns arise there, the study should be listed as 'awaiting classification' while clarifications (e.g. via contacting study authors) are sought. Only if concerns remain unresolved – or are severe – should exclusion be considered [62].

To apply the RIA tool, we checked the Retraction Watch Database for retraction notices. We assumed adequate ethical approval if the authors reported they had obtained ethical approval. We did not require the approval number because some studies were conducted in multiple centres and countries. We assumed that authorship was implausible if most study authors were affiliated with the manufacturer of the medication tested. We assessed methodological plausibility by considering whether the method used for participant randomisation was adequately described and would lead to a random allocation of the participants. We considered a lack of block randomisation as a cause for concern. We assumed that study results were implausible when the standardised mean difference (SMD) for pain intensity was large (Cohen's d > 0.8), given that all available drugs for the management of chronic neuropathic pain typically yield only small to moderate effects [84, 85, 86].

Two review authors (GA, WH) independently re-evaluated all trials included in the 2018 version of the review and assessed all newly eligible trials for research integrity. We resolved any discrepancies through discussion to reach consensus, involving a third author if necessary (LR) (for details, see Supplementary material 2; Supplementary material 8).

Strict application of the RIA tool indicated that we should have excluded 14 studies from analysis because they lacked prospective trial registration (Berman 2004 [87]; Ellis 2009 [88]; Frank 2008 [89]; Langford 2013 [90, 91]; NCT00710424 [92]; NCT01606176 [93]; NCT01606202 [94]; Nurmikko 2007 [95, 96]; Rog 2005 [97, 98]; Schimrigk 2017 [99]; Selvarajah 2010 [100]; Serpell 2014 [101, 102]; Svendsen 2004 [103]; Toth 2012 [104]), and one study for lack of ethical approval (Xu 2020 [105]). Authors of one study with potentially implausible results did not respond to our request for information and should have been listed as 'awaiting classification' (Seevathee 2025 [106]). Ultimately, strict application of the RIA criteria meant that only five studies would have been included in the analysis (D'Andre 2024 [107]; Eibach 2021 [108]; Hansen 2023 [109]; Lynch 2014 [110]; Zubcevic 2023 [111]).

We decided against a strict application of the criteria, both for general reasons to do with the evolution of clinical trial standards over time, and for specific reasons to do with trials in cannabis, as follows.

- Registration of clinical studies was not mandatory in the early 2000s, and requirements were introduced gradually and became increasingly stringent. For example, since 2004, the International Committee of Medical Journal Editors (ICMJE) has required prospective registration in an ICMJE-recognised registry as a condition for publication [112]. The 2013 revision of the Declaration of Helsinki further mandated that all clinical trials be registered in a publicly accessible database before enrolment begins [113]. In the European Union, registration and results reporting became legally binding with EU Regulation 536/2014 for pharmaceutical trials and EU Regulation 2017/745 for medical device studies [114].
- In Europe, mandatory ethics committee approval for drug trials was introduced in 2001 [115].
- Cannabis research was historically restricted in the USA, with the 1970 Controlled Substances Act classifying it as a Schedule I substance, defined as having 'no currently accepted medical use' and a 'high potential for abuse'. Consequently, researchers faced significant barriers to conducting studies, including: obtaining approval from multiple federal agencies; strict security requirements for storing and dispensing cannabis; and limited access to research-grade cannabis [116].

Taken together, the general limitations of studies conducted before 2010 and the specific challenges of cannabis-based research likely contributed to the historical evidence not meeting all proposed trustworthiness criteria. Therefore, we did not exclude studies which met our inclusion criteria but instead conducted a sensitivity analysis excluding studies that would have been excluded or listed as 'awaiting classification' by the RIA tool.

Data extraction and management

Three review authors (GA, PW, WH) independently extracted data using a standardised data extraction form, including details of the trial, participants, intervention, comparator, and outcomes. If necessary, we tried to obtain missing data by contacting study authors. At each step of data extraction, we resolved any discrepancies through discussion. We extracted the following information, if reported.

- Trial characteristics: setting and dates, inclusion/exclusion criteria, number of trial arms, treatment cross-overs, length of follow-up, attrition rates, funding, conflicts of interest, ethical approval, pre-registered study protocol; address and contact information of the corresponding author.
- Participant characteristics: numbers of participants screened, randomised, and in receipt of the intervention; type(s) of neuropathic pain conditions(s); age, gender, and pain intensity at baseline.
- Intervention: dose, frequency, duration and route of administration, co-therapies allowed, rescue medication.
- Control intervention: type of control, frequency, duration and route of administration, co-therapies allowed, rescue medication.
- Outcomes: as specified under Outcome measures.

Risk of bias assessment in included studies

Two review authors (GA, WH) independently assessed the risk of bias for each new study and checked the assessments of studies included in the first version of the review. We defined the following eight domains:

- random sequence generation (selection bias);
- allocation concealment (selection bias);
- blinding of participants and personnel (performance bias);
- blinding of outcome assessors (detection bias);
- incomplete outcome data (attrition bias);
- selective reporting (reporting bias);
- group similarity at baseline (selection bias);
- sample size (small-study bias).

Our approach adapts the original Cochrane tool (RoB 1) by enumerating the two final domains, which are usually grouped under the generic domain of 'other bias' [117], and reflects how risk of bias assessment was performed in a recent Cochrane review about cannabis-based medicines for chronic cancer pain [118]. See Supplementary material 9 for full details about how we applied the RoB tool.

We resolved any disagreements about the risk of bias judgements through discussion.

We classified studies as having an overall low risk of bias if they had zero to two domains rated as unclear or high risk; an overall unclear risk of bias if three to five domains were rated as unclear or high risk, and an overall high risk of bias if six to eight domains were rated as unclear or high risk [119].

Measures of treatment effect

We calculated numbers needed to treat for an additional beneficial outcome (NNTB) as the reciprocal of the absolute risk reduction (ARR) [120]. For unwanted effects, the NNTB becomes the number needed to treat for an additional harmful outcome (NNTH) and is calculated in the same manner. We used dichotomous data to calculate risk differences (RD) with 95% confidence intervals (CIs) using a fixed-effect model unless we found significant statistical or clinical heterogeneity (see Synthesis methods).

We calculated standardised mean differences (SMD) with 95% CIs for continuous variables using a fixed-effect model unless we found significant statistical or clinical heterogeneity.

We considered effect estimates statistically significant if the 95% CI for dichotomous outcomes did not include 1, or if the 95% CI for continuous outcomes did not include 0. A statistically significant effect does not necessarily mean that the estimated effect is clinically relevant. We assessed clinical relevance by defining a categorical outcome as demonstrating a clinically meaningful benefit or harm if the NNTB or NNTH was less than 10 [121]. We classified clinical effects based on NNTB/NNTH as follows: 7 to 9 indicated a small clinically relevant effect, 4 to 6 a moderate effect, and fewer than 4 a large clinically relevant effect. We used Cohen's benchmarks to interpret the magnitude of standardised mean differences (SMD) for continuous outcomes, with Hedges' g values of 0.2, 0.5, and 0.8 corresponding to small, medium, and large effects, respectively [122]. We labelled a g value of less than 0.2 to be a 'not substantial' effect size. We assumed a minimally important difference if Hedges' g value was 0.2 or greater [123].

Unit of analysis issues

The unit of analysis for this review was the randomised participant.

We split the control treatment arm between active treatment arms in a single study if the active treatment arms were not combined for analysis.

We included cross-over studies if separate data from the two periods were reported, if analyses showed no statistically significant carry-over effect, or if appropriate statistical adjustments were made in the presence of a significant carry-over effect.

Dealing with missing data

We used intention-to-treat (ITT) analysis, where the ITT population consisted of participants who were randomised, took at least one dose of the assigned study medication, and provided at least one post-baseline assessment.

Where means or standard deviations (SDs) were missing, we attempted to obtain these data by contacting trial authors. Where SDs were not available from trial authors, we calculated them from t values, P values, CIs, or standard errors reported by the studies [64]. Where rates of pain relief of 30% and 50% or greater were not reported or unavailable upon request, we calculated them from means and SDs using a validated imputation method [124].

Reporting bias assessment

We assessed publication bias using a method designed to detect the amount of unpublished data with a null effect required to make any result clinically irrelevant (usually taken to mean an NNTB of 10 or higher) [121].

Synthesis methods

We performed meta-analyses in RevMan using a random-effects model with the inverse variance method [125], due to substantial clinical heterogeneity arising from the inclusion of different neuropathic pain conditions. We estimated between-study heterogeneity using the DerSimonian and Laird method and calculated confidence intervals for the summary effect using the Wald-type approach, following current guidance in the *Cochrane Handbook for Systematic Reviews of Interventions* (Section 10.10.4.4) [64].

Investigation of heterogeneity and subgroup analysis

We assessed statistical heterogeneity visually [126], and using the I^2 statistic [127]. We declared statistical heterogeneity to be present if I^2 exceeded 40% (40% to 60%: moderate heterogeneity; 50% to 90%: substantial heterogeneity; 75% to 100%: considerable heterogeneity) [64].

As noted above, we did not plan subgroup analyses stratified by type of cannabis-based medicine.

Equity-related assessment

We did not assess health equity (absence of avoidable and unfair differences in health) in this review. By analysing the baseline characteristics of the study groups, we found that most studies did not report many health inequity characteristics (such as place of residence, culture/language, occupation, religion, education, socio-economic status, social capital, sexual orientation, and disability). We even found no subgroup analysis by age or gender in the studies analysed.

Sensitivity analysis

We performed a sensitivity analysis excluding studies which would have been excluded or classified as 'awaiting classification' by the RIA tool. We also conducted a post hoc sensitivity analysis excluding one outlier study and six studies with double-blind durations of less than four weeks, in response to feedback from one peer reviewer.

Certainty of the evidence assessment

Two review authors (GA, WH) independently rated the certainty of the evidence for the outcomes presented in the summary of findings tables, using the GRADE approach, which considers five domains: study limitations (risk of bias), inconsistency, imprecision, indirectness, and publication bias [128]. For each outcome, we categorised the certainty of the body of evidence as 'high', 'moderate', 'low', or 'very low'. We resolved any discrepancies through discussion or by consulting a third review author (LR). We reported the reasons for all decisions to downgrade the certainty of the evidence from the default 'high certainty' for evidence from RCTs.

We developed three summary of findings tables to present the main findings of three comparisons in a transparent and simple tabular format:

- THC-dominant medicines versus placebo;
- THC/CBD-balanaced medicines versus placebo;
- CBD-dominant medicines versus placebo.

The tables include key information concerning the certainty of evidence, the magnitude of effect of the interventions examined, and the sum of available data on the outcomes. The tables present the four critical outcomes and three important outcomes: participant-reported pain relief of 30% or greater, and the specific adverse events of nervous system disorders and psychiatric disorders.

Consumer involvement

The first version of the review did not involve consumers due to limited resources. One author (BL) of this review update is affected by neuropathic pain; she helped to affirm and select the outcomes of interest.

RESULTS

Description of studies

See Supplementary material 2; Supplementary material 3; Supplementary material 4, Supplementary material 5; Supplementary material 6.

Results of the search

The new search conducted on 29 January 2025 produced 2697 records: 2663 records from the database searches plus 34 from other sources. After removing 365 duplicates, 2332 records remained. We screened 2332 abstracts and titles and retrieved 15 full-text reports for further review. After reading the full reports, we excluded seven studies (see Excluded studies for reasons), and listed two studies as 'awaiting classification' (further details below). We included six new studies with 450 participants. We excluded Ware 2010 [129], which was included in the 2018 version of the review, because its dichotomous data could not be appropriately incorporated into our analyses due to the paired nature of the outcomes. With 15 studies involving 1737 participants from the 2018 review, this update included a total of 21 studies with 2187 participants in the qualitative and quantitative analysis (see Figure 1).

Figure 1. Study flow diagram

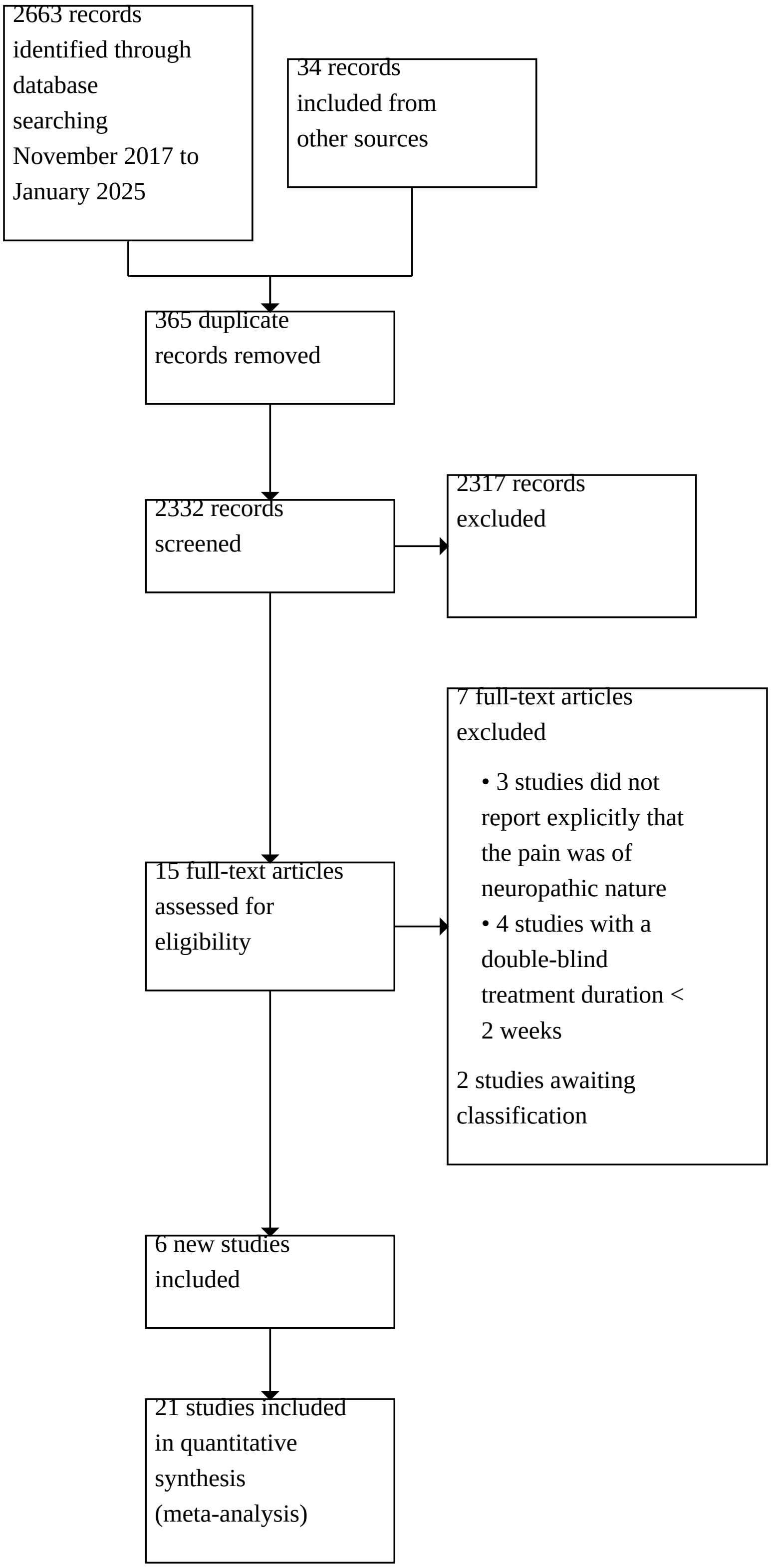


Figure 1. (Continued)

The protocols of four studies included in the 2018 review have been published in the time elapsed since we conducted the searches for that review (in November 2017) (Langford 2013; Nurmikko 2007; Rog 2005; Serpell 2014). These protocols included new data for the studies, which we included in the quantitative analysis. Three other studies included in the 2018 review (using data from trial register entries) have still not been published in peer-reviewed journals (NCT00710424; NCT01606176; NCT01606202).

Included studies

We summarise key study characteristics in Table 1 (study design; duration of randomised controlled period; setting; reference medication and comparator; type of neuropathic pain; study population). Detailed descriptions of the 21 included studies are available in Supplementary material 2, including information about study settings, authors' conflicts of interest, and rescue and co-medications used in the studies.

Study characteristics

Seven studies used a cross-over design and two an enriched-enrolment randomised withdrawal (EERW) design. One study started with a parallel-group design and subsequently re-randomised participants who responded to treatment into a withdrawal phase. We did not include the data of the withdrawal phase of this study in the analysis because analysing data from both phases introduces a unit-of-analysis issue. The remaining studies used a parallel-group design.

Three of 21 included studies were multi-armed studies (Berman 2004; Hansen 2023; Zubcevic 2023). Of these, two studies compared THC-dominant, CBD-dominant, and THC/CBD-balanced medicines to placebo (Hansen 2023; Zubcevic 2023). The third study compared THC-dominant and THC/CBD-balanced medicines to placebo (Berman 2004). The 18 remaining studies compared one study arm with cannabis-based medicine to a placebo arm.

Six studies had very short durations (two to three weeks), 12 were of short duration (four to 12 weeks), and three were of intermediate duration (more than 12 weeks to 26 weeks).

The sample sizes ranged from 18 to 339 participants. Attrition rates ranged from 0% to 33%.

Two studies were started after 2020, two studies were started between 2010 and 2020, and the remaining studies were conducted between 2001 and 2010.

Six studies were publicly funded, one study reported that there was no external funding, and the remaining studies were funded by the manufacturer of the drug.

Participant characteristics

Types of neuropathic pain

Five studies included participants with central neuropathic pain: four studies focused on participants with multiple sclerosis and one study on participants with multiple sclerosis or spinal cord injury. Fourteen studies included participants with peripheral neuropathic pain. Two studies included participants with central or peripheral pain of various aetiologies.

Four studies did not describe how neuropathic pain was diagnosed. Eight studies based the diagnosis on clinical examination. Three studies used clinical examination and predefined scores in neuropathic pain questionnaires. Four studies combined clinical examination with instrument-based diagnostic tests. Two studies used predefined scores in neuropathic pain questionnaires.

Demographics

The mean age of the participants ranged from 34 to 70 years. The percentage of women ranged from 0% to 90%.

Inclusion criteria

Thirteen studies required a pain score of 4 or higher on a 0-to-10 scale at baseline for inclusion. The remaining studies did not report on a defined pain intensity inclusion criterion. Five studies required that participants' pain was refractory to previous analgesics without specifying the type and dosage of previous unsuccessful analgesic therapy.

Exclusion criteria

All studies excluded people with major medical diseases (heart, liver, kidney, seizures). Most studies excluded people with previous or current mental disorders, and those with previous or current substance abuse. One study excluded people who had used marijuana in the month before study entry; two studies excluded people who had used cannabis in the previous three months without reporting data on cannabis use before this period. One study excluded participants if cannabinoids were detected on blood testing at study entry. One study stated that previous cannabis use was not an exclusion criterion.

Participants' previous experience of herbal cannabis

Ten studies reported the previous herbal cannabis experience of participants. Of these, only two studies reported medical and recreational use separately. The percentage of participants with previous herbal cannabis experience ranged from 7% to 91%. Of note, the rate of previous herbal cannabis experience was the highest in the study with inhaled medical cannabis, at 91% (Ellis 2009).

Intervention and comparator characteristics

THC-dominant

Two studies included a study arm with oral THC only (Hansen 2023; Zubcevic 2023). One study investigated inhaled (by pipe or cigarette) herbal cannabis without reporting CBD content (Ellis 2009). Three studies tested oral synthetic THC (nabilone, marinol) only (Frank 2008; Svendsen 2004; Toth 2012), and one study plant-derived THC (dronabinol) only (Schimrigk 2017). One study investigated a transdermal THC-dominant preparation which included low doses of CBD and cannabinol (CBN) (Seevathee 2025). The average daily dosages of oral THC ranged from 22.5 mg to 25 mg. The average daily oral dose of plant-derived dronabinol was 13 mg and of nabilone was 3 mg. Seevathee 2025 did not report on the average daily dose of a THC-dominant oral oil, which also included low doses of CBD and CBN.

THC/CBD-balanced

Nine studies tested an oromucosal spray (Langford 2013; Lynch 2014; NCT00710424; NCT01606176; NCT01606202; Nurmikko 2007; Rog 2005; Selvarajah 2010; Serpell 2014). The average use in the THC/CBD group ranged from 8 to 9 daily actuations (225 mg to 248 mg THC and 200 mg to 225 mg CBD). The average oral daily doses of oral THC/CBD were 15 mg/30 mg (Hansen 2023; Zubcevic 2023).

CBD-dominant

Two studies included a study arm with oral CBD only (Hansen 2023; Zubcevic 2023). One study investigated oral cannabidivarin (CBDV) (Eibach 2021). Two studies tested a CBD cream (D'Andre 2024; Xu 2020). The average daily dose of oral CBD was 45 mg (Hansen 2023) and 50 mg (Zubcevic 2023). The oral dosage was fixed with CBDV 400 mg/day in Eibach 2021. Two studies did not report on the average dose of CBD cream (D'Andre 2024; Xu 2020).

Comparators

All but one study compared cannabis-based medicines to placebo, with Frank 2008 using dihydrocodeine (DHC) as the comparator.

Excluded studies

The 2018 review identified 15 excluded studies (Abrams 2007 [130]; Corey-Bloom 2012 [131]; Karst 2003 [132]; Notcutt 2011 [133]; Novotna 2011 [134]; Rintala 2010 [135]; Turcotte 2015 [136]; Wade 2003 [137]; Wade 2004 [138]; Wallace 2015 [139]; Wilsey 2008 [140]; Wilsey 2013 [141]; Wissel 2006 [142]; Zajicek 2003 [143]; Zajicek 2012 [144]). The reasons for exclusion were: no explicit statement that participants' pain was neuropathic in nature (five studies); insufficient treatment duration (< two weeks' double-blind treatment duration; five studies); insufficient sample size (< 10 participants per treatment arm; two studies); ineligible participants (no pain or non-neuropathic pain; two studies); and ineligible outcome (one study).

In this review update, we excluded three studies in multiple sclerosis in which pain was a secondary outcome because they provided no explicit statement that the pain was of neuropathic nature (Collin 2010 [145]; Leocani 2015 [146]; Markova 2019 [147]), and four studies with an insufficient double-blind treatment duration (< two weeks) (Kittithamvongs 2025 [148]; NCT01037088 [149]; NCT03099005 [150]; Weizman 2024 [151]). As noted above, we reclassified Ware 2010, which was included in the 2018 version of the review, as an excluded study because the paired nature of the data did not allow the analysis of dichotomous outcomes.

Studies awaiting classification

The 2018 review identified three studies with unpublished results or of an unknown status, and we were unsuccessful in obtaining further information from authors. All three studies were investigations of nabilone by Canadian universities (NCT00699634 [152]; NCT01035281 [153]; NCT01222468 [154]). One was sponsored by the manufacturer of the drug (NCT00699634); the remaining two were university-funded (NCT01035281; NCT01222468). Our update searches in 2025 did not identify any new data posted in clinical trial registries for these studies. We identified two further studies as 'awaiting classification' (NCT02460692 [155]; van Amerongen 2018 [156]): both include the term 'neuropathic pain' in the study title, but do not report the methods used to diagnose neuropathic pain. The authors did not respond to requests for information.

Risk of bias in included studies

Using the original Cochrane risk of bias tool, we judged that five studies had an overall high risk of bias (NCT00710424; NCT01606176; NCT01606202; Selvarajah 2010; Xu 2020), 10 studies an overall unclear risk of bias (Berman 2004; Ellis 2009; Frank 2008; Langford 2013; Lynch 2014; Nurmikko 2007; Schimrigk 2017; Serpell 2014; Svendsen 2004; Toth 2012), and six studies an overall low risk (D'Andre 2024; Eibach 2021; Hansen 2023; Rog 2005; Seevathee 2025; Zubcevic 2023). Figure 2 presents a graph showing our risk of bias judgements as percentages across all included studies; Figure 3 depicts our judgements for each study and each risk of bias domain. See Supplementary material 2 for detailed information regarding the risk of bias assessment of each study.

Figure 2. Risk of bias graph: review authors' judgements about each risk of bias item presented as percentages across all included studies

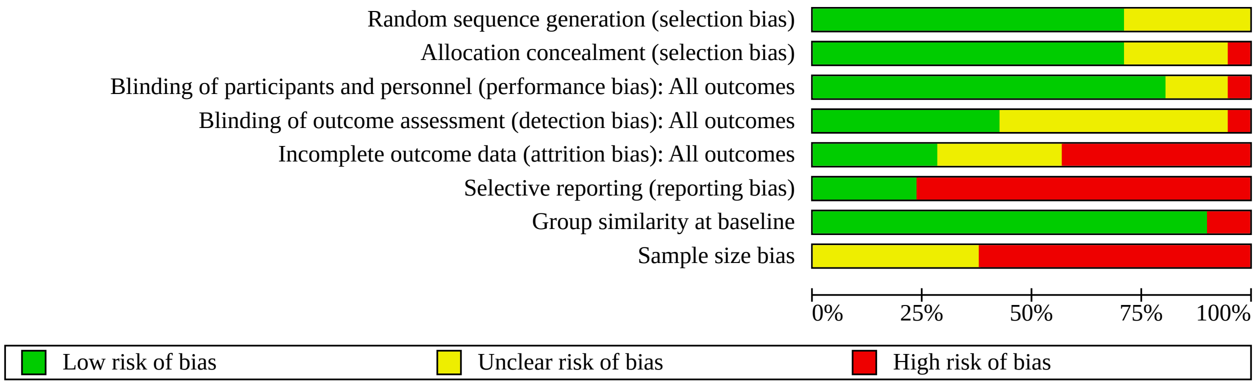
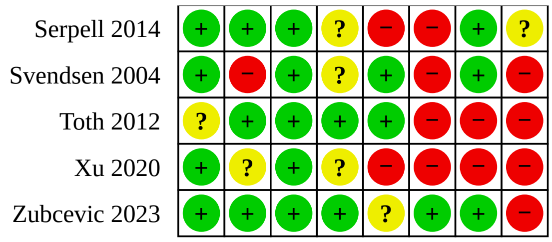


Figure 3. Risk of bias summary: review authors' judgements about each risk of bias item for each included study

	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias): All outcomes	Blinding of outcome assessment (detection bias): All outcomes	Incomplete outcome data (attrition bias): All outcomes	Selective reporting (reporting bias)	Group similarity at baseline	Sample size bias
Berman 2004	+	+	?	?	+	−	+	−
D'Andre 2024	+	+	+	+	−	+	+	−
Eibach 2021	+	+	+	+	?	+	+	−
Ellis 2009	+	+	?	?	−	−	+	−
Frank 2008	+	+	+	?	−	−	+	?
Hansen 2023	+	+	+	−	+	+	+	−
Langford 2013	+	+	−	?	?	−	+	?
Lynch 2014	?	+	+	+	?	−	+	−
NCT00710424	?	?	+	?	−	−	+	?
NCT01606176	?	?	+	?	?	−	+	−
NCT01606202	?	?	+	?	?	−	+	?
Nurmikko 2007	+	+	+	+	−	−	+	?
Rog 2005	+	+	+	+	+	−	+	−
Schimrigk 2017	+	+	+	+	−	−	+	?
Seevathee 2025	+	+	+	+	+	+	+	?
Selvarajah 2010	?	?	?	?	−	−	+	−
Serpell 2014	+	+	+	?	−	−	+	?

Figure 3. (Continued)



Random sequence generation

Fifteen studies reported their methods for random sequence generation and were therefore deemed to have a low risk of selection bias. We assessed the remaining six studies to have an unclear risk of bias in this domain as they did not adequately describe randomisation methods (Lynch 2014; NCT00710424; NCT01606176; NCT01606202; Selvarajah 2010; Toth 2012).

Allocation concealment

Allocation concealment was adequately described and therefore of low risk of bias in all but six studies. We judged five studies to have an unclear risk of bias in this domain due to inadequate descriptions (NCT00710424; NCT01606176; NCT01606202; Selvarajah 2010; Xu 2020), and one study to have a high risk of bias (Svendsen 2004).

Blinding of participants and personnel

Blinding of participants and personnel was adequately described and therefore of low risk of bias in all but four studies. We judged three studies to have an unclear risk of bias in this domain due to inadequate descriptions (Berman 2004; Ellis 2009; Selvarajah 2010), and one study to have a high risk of bias as the placebo was distinguishable from the intervention (Langford 2013).

Blinding of outcome assessors

We judged nine studies to have a low risk of detection bias as blinding of outcome assessment was adequately described (D'Andre 2024; Eibach 2021; Lynch 2014; Nurmikko 2007; Rog 2005; Schimrigk 2017; Seevathee 2025; Toth 2012; Zubcevic 2023), and one study to have a high risk of bias in this domain (Hansen 2023). The remaining studies did not report adequate information for judgement (unclear risk).

Incomplete outcome data

Only two studies conducted an intention-to-treat (ITT) analysis for the critical outcome of pain, using the baseline observation carried forward (BOCF) method (Hansen 2023; Svendsen 2004). Eight studies performed completer analysis for some outcomes (D'Andre 2024; Frank 2008; NCT00710424; NCT01606176; NCT01606202; Nurmikko 2007; Selvarajah 2010; Serpell 2014). The remaining studies performed ITT analysis, using the last observation carried forward (LOCF) method.

The attrition rates were low (< 10%) in six studies (Berman 2004; NCT01606202; Rog 2005; Seevathee 2025; Svendsen 2004; Toth 2012), moderate (10% to 20%) in nine studies (D'Andre 2024; Eibach 2021; Ellis 2009; Hansen 2023; Langford 2013; Lynch 2014; NCT01606176; Nurmikko 2007; Zubcevic 2023), and high (> 20%) in six studies (Frank 2008; NCT00710424; Schimrigk 2017; Selvarajah 2010; Serpell 2014; Xu 2020). We judged the risk of attrition bias to be low in six studies (Berman 2004; Hansen 2023; Rog 2005; Seevathee 2025; Svendsen 2004; Toth 2012), unclear in six studies (Eibach 2021; Langford 2013; Lynch 2014; NCT01606176; NCT01606202; Zubcevic 2023), and high in nine studies (D'Andre 2024; Ellis 2009; Frank 2008; NCT00710424; Nurmikko 2007; Schimrigk 2017; Selvarajah 2010; Serpell 2014; Xu 2020).

Selective reporting

We judged 16 studies to be at a high risk of selective reporting bias for the following reasons: we were unable to find study protocols for six studies (Berman 2004; Ellis 2009; Frank 2008; Nurmikko 2007; Svendsen 2004; Xu 2020); authors of one study reported on request that there was no study protocol (Selvarajah 2010); one study's protocol did not include outcomes (was empty) (Toth 2012); and eight studies published their protocols after the study concluded (Frank 2008; Langford 2013; NCT00710424; NCT01606176; NCT01606202; Rog 2005; Schimrigk 2017; Serpell 2014). The remaining five studies reported a prospective trial registration.

Group similarity at baseline

We judged all but two studies to have a low risk of bias because there were no significant differences in demographic and clinical variables at baseline; we rated Toth 2012 and Xu 2020 as high risk in this domain.

Sample size

We judged eight studies to have an unclear risk of sample size bias as they had 50 to 199 participants per study arm (Frank 2008; Langford 2013; NCT00710424; NCT01606202; Nurmikko 2007; Schimrigk 2017; Serpell 2014; Seevathee 2025). We judged the remaining 13 studies to have a high risk of bias in this domain as they had fewer than 50 participants per study arm (Berman 2004; D'Andre 2024; Eibach 2021; Ellis 2009; Hansen 2023; Lynch 2014; NCT01606176; Rog 2005; Selvarajah 2010; Svendsen 2004; Toth 2012; Xu 2020; Zubcevic 2023).

Synthesis of results

We included no studies for two comparisons of interest: THC/CBD-balanced medicines versus an active pharmacological intervention and CBD-dominant medicines versus an active pharmacological intervention.

In the results section, we report simple aggregates of the number of events in each treatment group along with the corresponding percentages for each outcome. These numbers may differ slightly from the absolute effects shown in the Summary of Findings tables. The Summary of Findings tables present the event rates in the intervention groups together with their confidence intervals based on the control group event rate and the risk difference. As described in the methods, any NNTs were calculated from the risk difference and not from the difference between the simple aggregated event rates in the intervention and control groups.

Comparison 1. THC-dominant medicines versus placebo

See Summary of findings 1 and Supplementary material 5.

Critical outcomes

We downgraded the certainty of evidence for all critical outcomes by one level each due to study design limitations (most studies were at an overall high or unclear risk of bias) and indirectness (most studies excluded people with mental disorders and major medical diseases).

Participant-reported pain relief of 50% or greater

We analysed seven studies with 534 participants. Ninety-six of 298 (32.2%) participants in the THC-dominant group and 55 of 236 (23.3%) participants in the placebo group reported pain relief of 50% or greater (risk difference (RD) 0.14, 95% CI -0.07 to 0.34; P = 0.19, I^2 = 89%). The certainty of evidence was very low, downgraded by one level each for risk of bias, indirectness, imprecision (CI included zero), and inconsistency (I^2 > 40%). There was no clear evidence for an effect.

Patient Global Impression of Change (PGIC) rating of 'much' or 'very much' improved

We analysed two studies with 72 participants. Fourteen of 45 (31.1%) participants in the THC-dominant group and five of 27 (18.5%) participants in the placebo group reported being much or very much improved (RD 0.17, 95% CI -0.24 to 0.58; P = 0.42, I^2 = 75%). The certainty of evidence was very low, downgraded by one level each for risk of bias, indirectness, imprecision (CI included zero), and inconsistency (I^2 > 40%). There was no clear evidence for an effect.

Withdrawals due to adverse events

We analysed six studies with 511 participants. Twenty-one of 289 (7.3%) participants in the THC-dominant group and six of 222 (2.7%) participants in the placebo group withdrew due to adverse events (RD 0.03, 95% CI -0.02 to 0.08; P = 0.22, I^2 =31%). The certainty of evidence was very low, downgraded by one level each for risk of bias, imprecision (CI included zero), and indirectness. There was no clear evidence for an effect.

Serious adverse events

We analysed seven studies with 537 participants. Twenty-one of 302 (7.0%) participants in the THC-dominant group and 10 of 235 (4.3%) participants in the placebo group reported serious adverse events (RD 0.02, 95% CI -0.01 to 0.06; P = 0.25, I^2 = 0%). The certainty of evidence was very low, downgraded by one level each for risk of bias, imprecision (CI included zero), and indirectness. There was no clear evidence for an effect.

Important outcomes

We did not assess the following outcomes using GRADE: mean pain intensity, health-related quality of life, sleep problems, psychological distress, withdrawal due to lack of efficacy, and any adverse event.

Participant-reported pain relief of 30% or greater

We analysed seven studies with 566 participants. In the THC-dominant group, 173 of 314 (55.1%) participants reported pain relief of 30% or greater, compared to 102 of 252 (40.5%) participants in the placebo group (RD 0.16, 95% CI -0.08 to 0.40; P = 0.20, I^2 = 88%). The certainty of evidence was very low, downgraded by one level each for risk of bias, indirectness, imprecision (95% CI included zero), and inconsistency (I^2 > 40%). There was no clear evidence for an effect.

Adverse events of the nervous system

We analysed five studies with 439 participants. In the THC-dominant group, 123 of 253 (48.6%) participants in the THC-dominant group reported adverse events of the nervous system, compared to 42 of 186 (22.6%) participants in the placebo group (RD 0.25, 95% CI 0.14 to 0.37; P < 0.001, I^2 = 26%). The number needed to treat for an additional harmful outcome (NNTH) was 4 (3 to 6). According to the predefined categories, there was a moderate clinically relevant harm of nervous system-related adverse events in THC-dominant medicine group. Using GRADE, the certainty of evidence was low, downgraded by one level each for risk of bias and indirectness. THC-dominant medicines may increase adverse events related to the nervous system.

Adverse events related to psychiatric disorders

We analysed four studies with 368 participants. Fifteen of 206 (7.3%) participants in the THC-dominant medicine group and three of 162 (1.9%) participants in the placebo group reported psychiatric adverse events (RD 0.01, 95% CI -0.01 to 0.03; P = 0.30, I^2 = 0%). The certainty of evidence was very low, downgraded by one level each for risk of bias, indirectness, and imprecision (95% CI included zero). There was no clear evidence for an effect.

Mean pain intensity

We analysed eight studies with 583 participants (standardised mean difference (SMD) -0.52, 95% CI -0.99 to -0.06; P = 0.03, I^2 = 83%). According to the predefined categories, there was a moderate clinically relevant benefit for THC-dominant medicines.

Health-related quality of life

We analysed four studies with 167 participants (SMD -0.36, 95% CI -1.14 to 0.42; P = 0.36, I^2 = 80%). There was no clear evidence for an effect.

Sleep problems

We analysed three studies with 143 participants (SMD -0.87, 95% CI -1.98 to 0.25; P = 0.13, I^2 = 87%). There was no clear evidence for an effect.

Psychological distress

We analysed four studies with 150 participants (SMD -0.13, 95% CI -0.47 to 0.20; P = 0.44, I^2 = 0%). There was no clear evidence for an effect.

Withdrawals due to lack of efficacy

We analysed five studies with 443 participants. Three of 255 (1.2%) participants in the THC-dominant medicine group and two of 188 (1.1%) participants in the placebo group withdrew due to lack of efficacy (RD 0.01, 95% CI -0.02 to 0.03; P = 0.53, I^2 = 0%). There was no clear evidence for an effect.

Any adverse event

We analysed five studies with 460 participants. In the THC-dominant medicine group, 169 of 243 (69.5%) participants reported adverse events, compared to 116 of 217 (53.5%) participants in the placebo group (RD 0.17, 95% CI 0.01 to 0.33; P = 0.04, I^2 = 75%). According to the predefined categories, there was a moderate clinically relevant harm from THC-dominant medicines.

Comparison 2. THC/CBD-balanced medicines versus placebo

See Summary of findings 2 and Supplementary material 5.

Critical outcomes

We downgraded the certainty of the evidence for all critical outcomes by one level for indirectness (most studies excluded people with mental disorders and major medical diseases).

Participant-reported pain relief of 50% or greater

We analysed eight studies with 746 participants. Ninety-six of 401 (23.9%) participants in the THC/CBD-balanced group and 61 of 345 (17.7%) participants in the placebo group reported pain relief of 50% or greater (RD 0.04, 95% CI 0.00 to 0.08; P = 0.08, I^2 = 0%). The certainty of evidence was very low, downgraded by one level each for risk of bias, indirectness, and imprecision (95% CI included zero). There was no clear evidence for an effect.

PGIC rating of 'much' or 'very much' improved

We analysed seven studies with 1145 participants. In the THC/CBD-balanced group, 151 of 586 (25.8%) participants reported being much or very much improved, compared to 110 of 559 (19.7%) participants in the placebo group (RD 0.07, 95% CI 0.02 to 0.11; P = 0.005, I^2 = 0%). The number needed to treat for an additional beneficial outcome (NNTB) was 16 (95% CI 9 to 50). According to the predefined categories, there was no clinically relevant effect on PGIC rating of 'much' or 'very much' improved for THC/CBD-balanced medicines. The certainty of evidence was low, downgraded by one level each for risk of bias and indirectness. THC/CBD-balanced medicines may slightly increase the number of participants with a PGIC rating of 'much' or 'very much' improved.

Withdrawals due to adverse events

We analysed 11 studies with 1449 participants. Ninety-six of 758 (12.7%) participants in the THC/CBD-balanced group and 47 of 691 (6.8%) participants in the placebo group withdrew due to adverse events (RD 0.05, 95% CI 0.02 to 0.09; P < 0.001, I^2 = 26%). The NNTH was 20 (95% CI 11 to 50). According to the predefined categories, there was no clinically relevant harm with regard to withdrawals due to adverse events compared to placebo. The certainty of evidence was low, downgraded by one level each for risk of bias and indirectness. THC/CBD-balanced medicines may slightly increase withdrawal rates due to adverse events.

Serious adverse events

We analysed 11 studies with 1449 participants. Forty-nine of 758 (6.5%) participants in the THC/CBD-balanced group and 38 of 691 (5.5%) participants in the placebo group reported serious adverse events (RD 0.01, 95% CI -0.02 to 0.03; P = 0.64, I^2 = 0%). The certainty of evidence was very low, downgraded by one level each for risk of bias, indirectness, and imprecision (95% CI included zero). There was no clear evidence for an effect.

Important outcomes

We did not assess the following outcomes using GRADE: mean pain intensity, health-related quality of life, sleep problems, psychological distress, withdrawals due to lack of efficacy, and any adverse event.

Participant-reported pain relief of 30% or greater

We analysed 10 studies with 1285 participants. In the THC/CBD-balanced group, 272 of 674 (40.4%) participants reported pain relief of 30% or greater, compared to 205 of 611 (33.6%) participants in the placebo group (RD 0.07, 95% CI 0.02 to 0.12; P = 0.009, I^2 = 0%). The NNTB was 16 (95% CI 8 to 50). According to the predefined categories, THC/CBD-balanced medicines conferred no clinically relevant benefit on participant-reported pain relief of 30% or greater. The certainty of evidence was low, downgraded by one level each for risk of bias and indirectness. THC/CBD-balanced medicines may increase the number of participants reporting pain relief of 30% or greater.

Adverse events of the nervous system

We analysed 11 studies with 1445 participants. In the THC/CBD-balanced medicine group, 465 of 755 (61.6%) participants reported adverse events of the nervous system, compared to 175 of 690 (25.4%) participants in the placebo group (RD 0.39, 95% CI 0.23 to 0.55; P < 0.001, I^2 = 91%). The NNTH was 3 (2 to 4). According to the predefined categories, THC/CBD-balanced medicines led to substantially more nervous system disorder adverse events than placebo. The certainty of evidence was very low, downgraded by one level each for risk of bias, indirectness, and inconsistency (I^2 > 40%).

Adverse events related to psychiatric disorders

We analysed nine studies with 1375 participants. Eighty-one of 709 (11.4%) participants in the THC/CBD-balanced medicines group and 18 of 666 (2.7%) participants in the placebo group reported psychiatric adverse events (RD 0.08, 95% CI 0.03 to 0.13; P = 0.001, I^2 = 66%). The NNTH was 12 (8 to 33). According to the predefined categories, THC/CBD-balanced medicines did not lead to a clinically relevant increase in psychiatric disorder adverse events compared to placebo. The certainty of evidence was very low, downgraded by one level each for risk of bias, indirectness, and inconsistency (I^2 > 40%). There was no clear evidence for an effect.

Mean pain intensity

We analysed 12 studies with 1390 participants (SMD -0.20, 95% CI -0.31 to -0.09; P < 0.001, I^2 = 3%). According to the predefined categories, there was a small clinically relevant benefit from THC/CBD-balanced medicines.

Health-related quality of life

We analysed 11 studies with 1348 participants (SMD -0.00, 95% CI -0.19 to 0.19; P = 1.00, I^2 = 59%). There was no clear evidence for an effect.

Sleep problems

We analysed nine studies with 1266 participants (SMD -0.20, 95% CI -0.33 to -0.06; P = 0.004, I^2 = 24%). According to the predefined categories, there was a small clinically relevant benefit from THC/CBD-balanced medicines.

Psychological distress

We analysed seven studies with 694 participants (SMD -0.06, 95% CI -0.26 to 0.15; P = 0.60, I^2 = 31%). There was no clear evidence for an effect.

Withdrawals due to lack of efficacy

We analysed nine studies with 1081 participants (RD -0.01, 95% CI -0.03 to 0.01; P = 0.38, I^2 = 0%). There was no clear evidence for an effect.

Any adverse event

We analysed eight studies with 1303 participants (RD 0.16, 95% CI 0.11 to 0.21; P < 0.001, I^2 = 13%). According to the predefined categories, there was a moderate clinically relevant harm from THC/CBD-balanced medicines.

Comparison 3. CBD-dominant medicines versus placebo

See Summary of findings 3 and Supplementary material 5.

Critical outcomes

We downgraded the certainty of the evidence for all critical outcomes by one level for indirectness (most studies excluded people with mental disorders and major medical diseases).

Participant-reported pain relief of 50% or greater

We analysed five studies with 208 participants. Fifteen of 122 (12.3%) participants in the CBD-dominant group and 18 of 86 (20.9%) participants in the placebo group reported pain relief of 50% or greater (RD -0.08, 95% CI -0.20 to 0.05; P = 0.23, I^2 = 45%). The certainty of evidence was very low, downgraded by one level each for indirectness, imprecision (95% CI included zero), and inconsistency (I^2 > 40%). There was no clear evidence for an effect.

PGIC rating of 'much' or 'very much' improved

We analysed two studies with 79 participants. Eight of 49 (16.3%) participants in the CBD-dominant group and six of 30 (20.0%) participants in the placebo group reported being much or very much improved (RD -0.03, 95% CI -0.22 to 0.16; P = 0.75, I^2 = 9%). The certainty of evidence was low, downgraded by one level each for indirectness and imprecision (95% CI included zero). CBD-dominant medicines may increase or decrease the number of participants reporting a PGIC rating of much or very much improved.

Withdrawals due to adverse events

We analysed five studies with 213 participants. Three of 126 (2.4%) participants in the CBD-dominant group and zero of 87 (20%) participants in the placebo group withdrew due to adverse events (RD 0.02, 95% CI -0.03 to 0.06; P = 0.48, I^2 = 0%). The certainty of evidence was low, downgraded by one level each for indirectness and imprecision (few events; 95% CI included zero). CBD-dominant medicines may increase or decrease withdrawals due to adverse events.

Serious adverse events

We analysed five studies with 213 participants. Two of 126 (1.6%) participants in the CBD-dominant group and zero of 87 (0%) participants in the placebo group reported serious adverse events (RD 0.02, 95% CI -0.03 to 0.06; P = 0.51, I^2 = 0%). The certainty of evidence was low, downgraded by one level each for indirectness and imprecision (few events, 95% CI included zero). CBD-dominant medicines may increase or decrease serious adverse events.

Important outcomes

We did not use the GRADE approach to assess the certainty of the evidence for the following outcomes: mean pain intensity, health-related quality of life, sleep problems, psychological distress, withdrawals due to lack of efficacy, and any adverse events.

Participant-reported pain relief of 30% or greater

We analysed five studies with 208 participants. Thirty-four of 122 (27.9%) participants in the CBD-dominant group and 27 of 86 (31.4%) participants in the placebo group reported pain relief of 30% or greater (RD -0.04, 95% CI -0.17 to 0.09; P = 0.55, I^2 = 35%). The certainty of evidence was low, downgraded by one level each for indirectness and imprecision (95% CI included zero). CBD-dominant medicines may increase or decrease participant-reported pain relief of 30% or greater.

Adverse events of the nervous system

We analysed five studies with 208 participants. Twenty-one of 123 (17.1%) participants in the CBD-dominant medicine group and 14 of 85 (16.5%) participants in the placebo group reported adverse events of the nervous system (RD -0.03, 95% CI -0.10 to 0.03; P = 0.32, I^2 = 4%). The certainty of evidence was low, downgraded by one level each for indirectness and imprecision (95% CI included zero). CBD-dominant medicines may increase or decrease adverse events of the nervous system.

Adverse events related to psychiatric disorders

We analysed five studies with 208 participants. Seven of 123 (5.7%) participants in the CBD-dominant medicine group and three of 85 (3.5%) participants in the placebo group reported psychiatric adverse events (RD -0.01, 95% CI -0.06 to 0.04; P = 0.71, I^2 = 0%). The certainty of evidence was low, downgraded by one level each for indirectness and imprecision (95% CI included zero). CBD-dominant medicines may increase or decrease adverse events related to psychiatric disorders.

Mean pain intensity

We analysed five studies with 173 participants (SMD 0.06, 95% CI -0.63 to 0.75; P = 0.87, I^2 = 78%). There was no clear evidence for an effect.

Health-related quality of life

We analysed two studies with 80 participants (SMD -0.23, 95% CI -0.70 to 0.23; P = 0.32, I^2 = 0%). There was no clear evidence for an effect.

Sleep problems

We analysed one study with 44 participants (SMD 0.24, 95% CI -0.41 to 0.89; P = 0.46). There was no clear evidence for an effect.

Psychological distress

We analysed one study with 34 participants (SMD -0.13, 95% CI -0.87 to 0.61, P = 0.72). There was no clear evidence for an effect.

Withdrawals due to lack of efficacy

We analysed five studies with 213 participants (RD 0.00, 95% CI -0.04 to 0.04; P = 0.89, I^2 = 0%). There was no clear evidence for an effect.

Any adverse event

We analysed four studies with 176 participants (RD 0.01, 95% CI -0.07 to 0.09; P = 0.81, I^2 = 0%). There was no clear evidence for an effect.

Comparison 4. THC-dominant medicines versus active pharmacological intervention

One study compared nabilone with dihydrocodeine (DHC) in 73 participants (Frank 2008). The certainty of the evidence for each critical outcome was low, downgraded for indirectness (the study excluded people with current or historical substance abuse and major medical diseases) and imprecision (only one study). We present a qualitative analysis of the study results in Supplementary material 10.

Subgroup analysis and investigation of heterogeneity

The heterogeneity (defined as I^2 > 40%) we detected in some outcomes might be due to clinical differences across the included studies, such as different methods of assessment and different types of neuropathic pain syndromes analysed.

Sensitivity analysis

Excluding studies that failed the Research Integrity Assessment (RIA) did not alter the direction or magnitude of effect estimates for the critical outcomes analysed in the THC-dominant, THC/CBD-balanced, or CBD-dominant comparisons. The only exception was the analysis of withdrawals due to adverse events in the THC/CBD-balanced medicine comparison, where the 95% confidence interval widened to include zero.

Removing the outlier study Seevathee 2025 did not change the direction or magnitude of the effect estimate for pain relief of 50% or greater in the analysis of THC-dominant medicines. This study did not report the other three critical outcomes. Removing studies with a double-blind duration of two or three weeks did not change the direction or magnitude of the effect estimates for the critical outcomes analysed in the THC-dominant and THC/CBD-balanced comparisons. The only exceptions were that the 95% confidence interval for withdrawals due to adverse events with THC-dominant medicines and for pain relief of 50% or greater with THC/CBD-balanced medicines no longer included zero (Supplementary material 11).

Equity assessment

We did not assess equity.

Reporting biases

We could not conduct the planned publication bias assessment because the observed NNTB values for all dichotomous benefit outcomes were already at least 10. Since the method requires an initial clinically relevant effect (NNTB < 10), it could not be applied.

DISCUSSION

Summary of main results

We included a total of 21 studies with 2187 participants in the qualitative and quantitative analysis, comprised of 15 studies involving 1737 participants identified for the 2018 review [2], and six new studies with 450 participants identified in this review update.

Six studies included participants with central neuropathic pain (multiple sclerosis, spinal cord injury); 13 studies included participants with peripheral neuropathic pain (plexus injury, peripheral neuropathy of different aetiologies, trigeminal neuropathy); and two studies included participants with both central and peripheral neuropathic pain.

Seven studies were with tetrahydrocannabinol (THC)-dominant medicines: five studies administered oral THC, one study administered a THC oromucosal spray, and one study administered a THC-dominant transdermal preparation which included low doses of cannabidiol (CBD) and cannabinol. Ten studies administered a balanced THC/CBD oromucosal spray and two an oral preparation with balanced THC and CBD. Two studies included a study arm with oral CBD. One study investigated oral cannabidivarin (similar to CBD) and two studies tested a CBD cream. All studies but one compared the intervention to placebo. One study compared the intervention to dihydrocodeine. The duration of the randomised period ranged from two to 26 weeks.

There is no clear evidence for an effect by THC-dominant medicines on pain relief of at least 50%, Patient Global Impression of Change (PGIC) rating of 'much' or 'very much' improved, withdrawals due to adverse events, and serious adverse events (very low-certainty evidence).

There is no clear evidence for an effect by THC/CBD-balanced medicines on pain relief of at least 50% and serious adverse events (very low-certainty evidence). THC/CBD-balanced medicines may increase the number of participants with a PGIC rating of 'much' or 'very much' improved and may increase withdrawals due to adverse events (low-certainty evidence).

There is no clear evidence for an effect by CBD-dominant medicines on pain relief of at least 50%. They may increase or decrease the number of participants with a PGIC rating of 'much' or 'very much' improved, serious adverse events, and withdrawals due to adverse events (low-certainty evidence).

Limitations of the evidence included in the review

Overall, the evidence was limited in both methodological rigour and clinical relevance.

Fourteen of the 22 studies were at high risk of bias because they were small. Two recent studies did not achieve their target sample sizes because the COVID-19 pandemic impeded recruitment (Hansen 2023; Zubcevic 2023). There are concerns about the influence of random chance due to the small amount of available data, as well as the risk of bias inherent in small studies, particularly in pain research [157, 158, 159, 160, 161]. Cochrane reviews have been criticised for perhaps over-emphasising results of underpowered studies or analyses [162, 163]. On the other hand, it may be unethical to ignore potentially important information from small studies or to randomise more participants if a meta-analysis including small, existing studies provided conclusive evidence.

We found the evidence for most outcomes to be low or very low because of limitations in study design, indirectness (people with major medical disorders were excluded, thus limiting the applicability of the evidence to routine clinical care), and inconsistent results.

Seven studies reviewed used a cross-over design with a study duration of between two and three weeks for each period. Cross-over designs have methodological issues that could lead to bias [164]. The short study duration limits their applicability. In addition, there are issues concerning the time needed (if any) for washout between treatment periods [1].

The European Medicines Agency recommends that study duration for chronic neuropathic pain trials should be at least 12 weeks after a stable dose is achieved in order to exclude a transient effect [165]. Only four studies included in this review lasted at least 12 weeks.

Most studies selected statistical methods (last observation carried forward, completer analysis) that bias results towards exaggerating the efficacy of drugs [1].

Some studies reported only some selected adverse events related to the nervous system or psychiatric disorders. Therefore, we might have underestimated the frequency of these adverse events.

Some studies did not report on participants' previous cannabis use; the ones which reported did not distinguish between recreational and medical cannabis use. Where reported, the number of participants with previous cannabis use ranged from 7% to 91%. No subgroup comparisons (former cannabis users versus cannabis-naïve participants) were conducted by any study. A prospective observational study found that the rate of non-serious adverse events among current cannabis users was lower than that among ex-cannabis users or cannabis-naïve users [166]. Therefore, we do not know if the review's results on benefits, tolerability, and harms of the RCTs reviewed are valid for cannabis-naïve participants.

People with chronic neuropathic pain exhibit a variety of pain-related sensory symptoms [167]. They use different descriptors for their pain (e.g. burning, tugging, pricking, cramping). None of the neuropathic pain scales available covers all potential descriptors of neuropathic pain [168]. Five of the newly included studies used and reported the results of combined neuropathic pain scales. The different instruments and descriptors of neuropathic pain used did not allow a quantitative analysis of single descriptors of neuropathic pain. Therefore, we do not know the efficacy of cannabis-based medicines for specific qualities of neuropathic pain.

Limitations of the review processes

The absence of publication bias (unpublished trials showing no benefit of cannabis-based medicines over placebo) can never be proved. We carried out a broad search for studies and feel it is unlikely that significant amounts of relevant data remain unknown to us.

We might have overestimated the risk of bias of some studies that did not report some details of methodology (e.g. randomisation and blinding procedures) and did not provide these details on request.

In the first version of the review, we found three industry-sponsored studies of THC/CBD spray with negative results, which have not been fully published yet. We also found three studies of nabilone, but the results were unknown; the study authors did not respond to our requests. For this review update, we identified two studies for which we could not reach a decision about inclusion or exclusion due to insufficient information, and we were unsuccessful in obtaining information from authors. Some authors of the included studies did not provide data on request that we required for quantitative synthesis and risk of bias assessment. Despite strengthened requirements for trial registration, full access to clinical trial data remains elusive [169]. Further research is (very) likely to have an important impact on our confidence in the estimates of effect and is likely to change the estimates.

The influence of allowed co-interventions (e.g. rescue medication) on positive effects and adverse events was unclear because the types and dosage of co-interventions were not clearly reported nor controlled for.

This systematic review included 2187 participants. To capture rare and potentially severe adverse events, a larger data set would have been necessary. For example, to capture an adverse event with a frequency of 1:100,000, observations from 300,000 participants would have been necessary [170].

Agreements and disagreements with other studies or reviews

During our updated search, we found more new systematic reviews on cannabis-based medicines for chronic neuropathic pain than new randomised controlled trials testing those medicines. These reviews tended to lump all cannabis-based medicines together in their analyses. A comparison of our review with most of these reviews was thus limited by the different approaches taken to analysis, as well as different inclusion criteria (study duration, number of participants, reliability of diagnosis of neuropathic pain, types of cannabis-based medicines included). We compare our results only with those of reviews providing results stratified by the ingredients of cannabis-based medicines.

We cannot share the optimistic conclusion of one review that inhaled cannabis is effective, well-tolerated, and safe in the treatment of chronic neuropathic pain [55]. The authors performed an individual participant data analysis of 178 participants from five studies of inhaled cannabis which reported the THC, but not the CBD content of the cannabis flower. They calculated an NNTB of 6 (95% CI 3 to 14) for a more than 30% reduction in pain scores compared to placebo. Withdrawals due to adverse events were found to be rare. The differences in our conclusions on the benefits and harms of THC-dominant medicines in chronic neuropathic pain can be explained by the exclusion of three one-day studies from our review (Abrams 2007; Wilsey 2008; Wilsey 2013), which the authors of the other review and meta-analysis included [55].

A 2022 systematic review by McDonagh and colleagues included 18 RCTs and seven cohort studies; 56% of these studies enroled people with neuropathic pain [73]. Two studies in its THC-dominant group (Schimrigk 2017; Toth 2012) and five studies in its THC/CBD-balanced group (Langford 2013; Lynch 2014; Rog 2005; Selvarajah 2010; Serpell 2014) were included in our review, too. The authors found a largely increased risk of dizziness for both types of cannabis-based medicines, corresponding to our findings on increased rates of central nervous system-related adverse events for both THC-dominant and THC/CBD-balanced medicines. McDongah and colleagues found THC-dominant and THC/CBD-balanced medicines to be effective in reducing pain [73]. In contrast, we found that the effects of THC-dominant and THC/CBD-balanced medicines on substantial pain relief (≥ 50%) are very uncertain. The effect size of THC/CBD-balanced medicines for moderate (≥ 30%) pain relief was not clinically relevant in our review, with an NNTB of 16. The two Danish trials with negative results in their THC and THC/CBD arms, which were not included by McDongah and colleagues [73], explain the divergence in the results for the efficacy of THC-dominant medicines.

We concluded that CBD has no effect on the relief of neuropathic pain. A 2024 systematic review by Moore and colleagues included 18 RCTs investigating CBD for any chronic pain [171], including three studies in CBD which were also included in this review (Hansen 2023; Xu 2020; Zubcevic 2023). Moore and colleagues concluded that CBD is not effective for pain relief. In contrast to Moore and colleagues [171], we did not find increased rates of serious adverse events. This divergent finding might be due to the different inclusion criteria of the two reviews.

AUTHORS' CONCLUSIONS

Implications for practice

There is no high-certainty evidence about the benefits and harms of any cannabis-based product, including herbal cannabis (marijuana), for adults with chronic neuropathic pain. The effects of tetrahydrocannabinol (THC)-dominant medicines on participant-reported pain relief of 50% or greater, Patient Global Impression of Change (PGIC) rating of 'much' or 'very much' improved, withdrawals due to adverse events, and serious adverse events are very uncertain. The effects of tetrahydrocannabinol/cannabidiol (THC/CBD)-balanced medicines on participant-reported pain relief of 50% or greater and serious adverse events are very uncertain. THC/CBD-balanced medicines may increase the number of participants with a PGIC rating of 'much' or 'very much' improved and the number of withdrawals due to adverse events. The effects of CBD-dominant medicines on participant-reported pain relief of 50% or greater are very uncertain. CBD-dominant medicines may increase or decrease the number of participants with a PGIC rating of 'much' or 'very much' improved, serious adverse events, and withdrawals due to adverse events.

A 2021 International Association for the Study of Pain (IASP) taskforce did not endorse the use of cannabis-based medicines for any chronic pain because of the lack of high-certainty evidence [58]. However, it did not recommend against their use. Some current clinical guidelines and position papers consider cannabis-based medicines as third- or fourth-line therapy for chronic neuropathic pain syndromes if established therapies (e.g. anticonvulsants, antidepressants) have not been effective, have not been tolerated, or are contraindicated [172; 51]. The Neuropathic Pain Special Interest Group of the IASP recommended against the use of cannabis-based medicines [86], basing its position on a systematic review that pooled all types of cannabis-based medicines in one quantitative analysis [86]. That review included 18 of the 21 studies included in this review.

There is substantial international variation in the availability of cannabis-based medicines, their regulatory approval for chronic pain, and the extent to which they are reimbursed by health-insurance systems [41, 53]. These factors should be kept in mind when considering the use of cannabis-based medicines in people with neuropathic pain.

Implications for research

General implications

The potential importance of combinations of CBD and THC with other cannabinoids (such as cannabinol or defined terpenes/flavonoids) – the so-called 'entourage effect' – still needs to be determined.

The development of peripherally acting cannabinoid agonists may reduce central nervous system and psychiatric adverse events.

The involvement of people with chronic pain in the planning (for example, by helping to define patient-relevant outcomes) and execution of clinical trials is critical, not only to their success, but also so that the participating communities can benefit from them [173].

GRADE-based implications

Risk of bias. There is a need for methodologically better designed and executed studies. Prospective cohort studies with initial randomisation and pragmatic designs should complement randomised controlled trials (RCTs). A minimum study duration of three months is recommended. Three-arm studies (study drug; standard drug treatment such as duloxetin or pregabalin; placebo) are desirable, in order to allow the comparative assessment of benefits and harms. Reporting the details of the assessment of adverse events (spontaneous reports, open questions, symptom questionnaires) is mandatory because the type and frequency of adverse events is influenced by the modes of assessment [174]. Adverse events have to be reported using the International Council for Harmonisation guidelines, and coded within organ classes using the Medical Dictionary for Regulatory Activities [79]. It is desirable that regulatory agencies standardise the assessment strategies of adverse events in RCTs.

Inconsistency. Neuropathic pain conditions should be investigated separately because different conditions can come with different placebo response rates and different effect sizes for the same intervention [175]. Details about how neuropathic pain was diagnosed should be reported. Adhering to the distinction drawn in the grading system developed by the IASP Special Interest Group on Neuropathic Pain between 'probable' and 'definite' neuropathic pain is desirable [176]. We recommend using validated neuropathic pain scales and reporting the effects of cannabis-based medicines on all items of these scales. In addition, subgrouping participants with neuropathic syndromes based on sensory profiles is possible and may be useful in clinical trial design to enrich the study population for treatment responders [167].

Indirectness. People with frequent comorbidities, such as relevant internal diseases and mental disorders, should be included in the studies.

Imprecision. Studies should include at least 150 participants per treatment arm.

Publication bias. We have found registered studies whose status and results are unknown. The registration of a study in a clinical register should include the obligation to provide information about the status of the study and publish the results after its completion.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: 10.1002/14651858.CD012182.pub3.

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Characteristics of studies awaiting classification

Supplementary material 5 Analyses

Supplementary material 6 Data package

Supplementary material 7 Terminology

Supplementary material 8 Research integrity assessment by Research Integrity Assessment (RIA) tool

Supplementary material 9 Risk of bias assessment

Supplementary material 10 THC-dominant medicines versus active pharmacological intervention

Supplementary material 11 Sensitivity analyses

ADDITIONAL INFORMATION

Acknowledgements

The protocol for this review followed the agreed template for neuropathic pain, which was developed collaboratively by Cochrane Pain, Palliative and Supportive Care (PaPaS), Cochrane Musculoskeletal, and Cochrane Neuromuscular Diseases.

The first version of the review was supported by the National Institute for Health Research (NIHR), via Cochrane Infrastructure funding to Cochrane Pain, Palliative and Supportive Care (PaPaS). The views and opinions expressed therein were those of the review authors and did not necessarily reflect those of the Systematic Reviews Programme, NIHR, the National Health Service (NHS), or the Department of Health.

We are grateful to the following Cochrane editorial staff who contributed to the revised review and provided methodological support: Lindsay Robertson, Evidence Synthesis Development Editor; Roses Parker, Commissioning Editor; and Steph Grohmann, Research Integrity Editor.

The following people conducted the editorial process for this review:

- Sign-off Editor (final decision): Neil O' Connell, Brunel University of London, United Kingdom;
- Managing Editor (selected peer reviewers, provided editorial guidance to authors, edited the article): Justin Mann, Cochrane Central Editorial Service;
- Editorial Assistant (conducted editorial policy checks, selected peer reviewers, collated peer reviewers' comments and supported the editorial team): Cynthia Stafford, Cochrane Central Editorial Service;
- Copy Editor (copy editing and production): Faith Armitage, Cochrane Central Production Service;
- Peer reviewers (provided comments and recommended an editorial decision): Xavier Moisset, Universite Clermant Auvergne, CHU Clermant-Ferrand, Inserm, Neuro-Dol, Clermant-Ferrand, France (clinical/content review); Dr Nan Nitra Than, former Professor of Community Medicine, Manipal University College Malaysia (clinical/content review); Brian Duncan (public and patient review); Jo-Ana Chase, Evidence Production and Methods Directorate (methods review); Jo Platt, Cochrane Editorial Information Specialist (search review).

Contributions of authors

Winfried Häuser (WH) conceived the review.

Frank Petzke and Winfried Häuser drafted the protocol for the first version of the review [177].

Winfried Häuser and Petra Klose developed the search strategy for this review update, which was based on the search strategy of the first version of the review developed by Joanne Abbott (PaPaS Information Specialist). Petra Klose conducted the updated searches.

Frank Petzke, Martin Mücke, and Winfried Häuser selected studies for inclusion for the first version of the review; Patrick Welsch and Winfried Häuser performed these tasks for the review update.

Frank Petzke, Martin Mücke, and Winfried Häuser extracted study data and assessed risk of bias for the first version of the review; Gülay Ateş and Winfried Häuser performed these tasks for the review update.

Frank Petzke and Winfried Häuser entered data and carried out the analysis for the first version of the review; Patrick Welsch and Winfried Häuser performed these tasks for the review update.

Gülay Ateş and Winfried Häuser drafted the manuscript of the updated review. Gülay Ateş, Tudor Philipps, Britta Lambers, Petra Klose, Lukas Radbruch, Patrick Welsch, and Winfried Häuser contributed to writing and revising the final report of the updated review.

Declarations of interest

Gülay Ateş: none.

Patrick Welsch: none. PW is a specialist pain physician and manages patients with neuropathic pain.

Petra Klose: none.

Tudor Phillips: none. TP is a specialist pain physician and manages patients with neuropathic pain.

Britta Lambers: none. BL is affected by neuropathic pain. She is vice-president of UVSD SchmerzLOS e.V. – Independent Association of Pain Patients in Germany.

Winfried Häuser: WH is a specialist in general internal medicine, psychosomatic medicine, and pain medicine, who manages patients with chronic neuropathic pain. He has received honoraria for consultation or educational lectures by Biotechnology fund, ECOMED, Global Life Science, Guidepoint Global, Otto Bock, Science Direct, University Essen-Duisburg, VIDAL, UCB, all not related to the submitted work. He has received one honorarium for reviewing a report entitled 'The efficacy and safety of medicinal cannabis in adult populations: an evidence review' by the Health Research Board of Ireland, and honoraria for textbook chapters on pain medicine by Ecomed. He is co-author of an industry-sponsored study, 'Full spectrum cannabis sativa DKJ 127 for chronic low back pain: a phase 3 randomized controlled trial', published in September 2025. He has received no honoraria for his authorship.

Lukas Radbruch: None. LR is a specialist in palliative care who treats patients with chronic neuropathic pain.

Sources of support

Internal sources

- Technische Universität München, Germany

  General institutional support

External sources

- The National Institute for Health Research (NIHR), UK

  For the first version of the review: NIHR Cochrane Programme Grant: 13/89/29 - Addressing the unmet need of chronic pain: providing the evidence for treatments of pain

Registration and protocol

Protocol first published: Mücke M, Phillips T, Radbruch L, Petzke F, Häuser W. Cannabinoids for chronic neuropathic pain. Cochrane Database of Systematic Reviews 2016, Issue 5. Art. No.: CD012182. DOI: 10.1002/14651858.CD012182.
Review first published: Mücke M, Phillips T, Radbruch L, Petzke F, Häuser W. Cannabis-based medicines for chronic neuropathic pain in adults. Cochrane Database of Systematic Reviews 2018, Issue 3. Art. No.: CD012182. DOI: 10.1002/14651858.CD012182.pub2.

Data, code and other materials

As part of the published Cochrane review, the following materials are made available for download by users of the Cochrane Library: full search strategies for each database (Supplementary material 1); full citations of each unique report for all studies included (Supplementary material 2), excluded studies (Supplementary material 3), and studies awaiting classification (Supplementary material 4); consensus risk of bias assessments (Supplementary material 5); and analysis data, including overall estimates and settings, subgroup estimates, and individual data rows (Supplementary material 6).

Notes

Assessed for updating in 2020

A restricted search in June 2020 did not identify any potentially relevant studies likely to change the conclusions. Therefore, this review has now been stabilised following discussion with the authors and editors. The review will be assessed for updating in two years. If appropriate, we will update the review before this date if new evidence likely to change the conclusions is published, or if standards change substantially, necessitating major revisions.

What's new

Date	Event	Description
23 January 2026	Amended	A sentence was added to the Results section to clarify how aggregate data are reported.

History

Protocol first published: Issue 5, 2016
Review first published: Issue 3, 2018

Date	Event	Description
19 January 2026	New citation required and conclusions have changed	New search performed: January 2025
19 January 2026	New search has been performed	New citation: Conclusions changed
20 July 2020	Amended	Minor error corrected in search strategy.

REFERENCES

1. Moore A, Derry S, Eccleston C, Kalso E. Expect analgesic failure; pursue analgesic success. *BMJ (Clinical Research Ed.)* 2013;**346**:f2690. [DOI: 10.1136/bmj.f2690]

2. Mücke M, Phillips T, Radbruch L, Petzke F, Häuser W. Cannabis-based medicines for chronic neuropathic pain in adults. *Cochrane Database of Systematic Reviews* 2018, Issue 3. Art. No: CD012182. [DOI: 10.1002/14651858.CD012182.pub2]

3. Jensen TS, Baron R, Haanpää M, Kalso E, Loeser JD, Rice AS, et al. A new definition of neuropathic pain. Pain 2011;**152**(10):2204-5. [DOI: 10.1016/j.pain.2011.06.017]

4. Demant DT, Lund K, Vollert J, Maier C, Segerdahl M, Finnerup NB, et al. The effect of oxcarbazepine in peripheral neuropathic pain depends on pain phenotype: a randomised, double-blind, placebo-controlled phenotype-stratified study. *Pain* 2014;**155**(11):2263-73. [DOI: 10.1016/j.pain.2014.08.014]

5. Helfert SM, Reimer M, Höper J, Baron R. Individualized pharmacological treatment of neuropathic pain. *Clinical Pharmacology and Therapeutics* 2015;**97**(2):135-42. [DOI: 10.1002/cpt.19]

6. Von Hehn CA, Baron R, Woolf CJ. Deconstructing the neuropathic pain phenotype to reveal neural mechanisms. *Neuron* 2012;**73**(4):638-52. [DOI: 10.1016/j.neuron.2012.02.008]

7. Baron R, Wasner G, Binder A. Chronic pain: genes, plasticity, and phenotypes. *Lancet Neurology* 2012;**11**(1):19-21. [DOI: 10.1016/S1474-4422(11)70281-]

8. Calvo M, Dawes JM, Bennett DL. The role of the immune system in the generation of neuropathic pain. *Lancet Neurology* 2012;**11**(7):629-42. [DOI: 10.1016/S1474-4422(12)70134-5]

9. Scholz J, Finnerup NB, Attal N, Aziz Q, Baron R, Bennett MI, et al. Classification Committee of the Neuropathic Pain Special Interest Group (NeuPSIG). The IASP classification of chronic pain for ICD-11: chronic neuropathic pain. *Pain* 2019;**160**:53-9. [DOI: 10.1097/j.pain.0000000000001365]

10. Van Hecke O, Austin SK, Khan RA, Smith BH, Torrance N. Neuropathic pain in the general population: a systematic review of epidemiological studies. *Pain* 2014;**155**:654-62.

11. Moore RA, Derry S, Taylor RS, Straube S, Phillips CJ. The costs and consequences of adequately managed chronic non-cancer pain and chronic neuropathic pain. *Pain Practice* 2014;**14**(1):79-94. [DOI: 10.1111/papr.12050]

12. Gustorff B, Dorner T, Likar R, Grisold W, Lawrence K, Schwarz F, et al. Prevalence of self-reported neuropathic pain and impact on quality of life: a prospective representative survey. *Acta Anaesthesiologica Scandinavica* 2008;**52**:132-6.

13. Bouhassira D, Lantéri-Minet M, Attal N, Laurent B, Touboul C. Prevalence of chronic pain with neuropathic characteristics in the general population. *Pain* 2008;**136**:380-7.

14. Torrance N, Smith BH, Bennett MI, Lee AJ. The epidemiology of chronic pain of predominantly neuropathic origin. Results from a general population survey. *Journal of Pain* 2006;**7**:281-9.

15. Hall GC, Carroll D, McQuay HJ. Primary care incidence and treatment of four neuropathic pain conditions: a descriptive study, 2002-2005. *BMC Family Practice* 2008;**9**:26.

16. Katusic S, Williams DB, Beard CM, Bergstralh EJ, Kurland LT. Epidemiology and clinical features of idiopathic trigeminal neuralgia and glossopharyngeal neuralgia: similarities and differences, Rochester, Minnesota, 1945-1984. *Neuroepidemiology* 1991;**10**:276-81.

17. Rappaport ZH, Devor M. Trigeminal neuralgia: the role of self-sustaining discharge in the trigeminal ganglion. *Pain* 1994;**56**:127-38.

18. Koopman JS, Dieleman JP, Huygen FJ, De Mos M, Martin CG, Sturkenboom MC. Incidence of facial pain in the general population. *Pain* 2009;**147**:122-7.

19. McQuay HJ, Smith LA, Moore RA. Chronic pain. In: Stevens A, Raftery J, Mant J, Simpson S, Editor(s). Health Care Needs Assessment, 3rd Series. Radcliffe Publishing, 2007.

20. Berger A, Dukes EM, Oster G. Clinical characteristics and economic costs of patients with painful neuropathic disorders. *Journal of Pain* 2004;**5**(3):143-9. [DOI: 10.1016/j.jpain.2003.12.004]

21. Berger A, Toelle T, Sadosky A, Dukes E, Edelsberg J, Oster G. Clinical and economic characteristics of patients with painful neuropathic disorders in Germany. *Pain Practice* 2009;**9**(1):8-17. [DOI: 10.1111/j.1533-2500.2008.00244.x]

22. Berger A, Sadosky A, Dukes E, Edelsberg J, Oster G. Clinical characteristics and patterns of healthcare utilization in patients with painful neuropathic disorders in UK general practice: a retrospective cohort study. *BMC Neurology* 2012;**12**:8.

23. Scott FT, Johnson RW, Leedham-Green M, Davies E, Edmunds WJ, Breuer J. The burden of herpes zoster: a prospective population based study. *Vaccine* 2006;**24**(9):1308-14. [DOI: 10.1016/j.vaccine.2005.09.026]

24. Kalso E, Aldington DJ, Moore RA. Drugs for neuropathic pain. *BMJ (Clinical Research Ed.)* 2013;**347**:f7339.

25. Moore RA, Chi CC, Wiffen PJ, Derry S, Rice AS. Oral nonsteroidal anti-inflammatory drugs for neuropathic pain. *Cochrane Database of Systematic Reviews* 2015, Issue 10. Art. No: CD010902. [DOI: 10.1002/14651858.CD010902.pub2]

26. Derry S, Moore RA. Topical capsaicin (low concentration) for chronic neuropathic pain in adults. *Cochrane Database of Systematic Reviews* 2012, Issue 9. Art. No: CD010111. [DOI: 10.1002/14651858.CD010111]

27. Derry S, Wiffen PJ, Moore RA, Quinlan J. Topical lidocaine for neuropathic pain in adults. *Cochrane Database of Systematic Reviews* 2014, Issue 7. Art. No: CD010958. [DOI: 10.1002/14651858.CD010958.pub2]

28. Derry S, Rice AS, Cole P, Tan T, Moore RA. Topical capsaicin (high concentration) for chronic neuropathic pain in adults. *Cochrane Database of Systematic Reviews* 2017, Issue 1. Art. No: CD007393. [DOI: 10.1002/14651858.CD007393.pub4]

29. Lunn MP, Hughes RA, Wiffen PJ. Duloxetine for treating painful neuropathy, chronic pain or fibromyalgia. *Cochrane Database of Systematic Reviews* 2014, Issue 1. Art. No: CD007115. [DOI: 10.1002/14651858.CD007115.pub3]

30. Wiffen PJ, Derry S, Bell RF, Rice AS, Tölle TR, Phillips T, et al. Gabapentin for chronic neuropathic pain in adults. *Cochrane Database of Systematic Reviews* 2017, Issue 6. Art. No: CD007938. [DOI: 10.1002/14651858.CD007938.pub4]

31. Moore RA, Derry S, Aldington D, Cole P, Wiffen PJ. Amitriptyline for neuropathic pain in adults. *Cochrane Database of Systematic Reviews* 2015, Issue 7. Art. No: CD008242. [DOI: 10.1002/14651858.CD008242.pub3]

32. Sultan A, Gaskell H, Derry S, Moore RA. Duloxetine for painful diabetic neuropathy and fibromyalgia pain: systematic review of randomised trials. *BMC Neurology* 2008;**8**:29. [DOI: 10.1186/1471-2377-8-]

33. Moore RA, Straube S, Wiffen PJ, Derry S, McQuay HJ. Pregabalin for acute and chronic pain in adults. *Cochrane Database of Systematic Reviews* 2009, Issue 3. Art. No: CD007076. [DOI: 10.1002/14651858.CD007076.pub2]

34. Moore RA, Cai N, Skljarevski V, Tölle TR. Duloxetine use in chronic painful conditions - individual patient data responder analysis. *European Journal of Pain (London, England)* 2014;**18**(1):67-75. [DOI: 10.1002/j.1532-2149.2013.00341.x]

35. Wiffen PJ, Derry S, Moore RA, Aldington D, Cole P, Rice AS, et al. Antiepileptic drugs for neuropathic pain and fibromyalgia - an overview of Cochrane reviews. *Cochrane Database of Systematic Reviews* 2013, Issue 11. Art. No: CD010567. [DOI: 10.1002/14651858.CD010567.pub2]

36. Moisset X. Neuropathic pain: evidence based recommendations. *La Presse Médicale* 2024;**53**:104232. [DOI: 10.1016/j.lpm.2024.104232]

37. Gaskell H, Derry S, Stannard C, Moore RA. Oxycodone for neuropathic pain in adults. *Cochrane Database of Systematic Reviews* 2016, Issue 7. Art. No: CD010692. [DOI: 10.1002/14651858.CD010692.pub3]

38. Stannard C, Gaskell H, Derry S, Aldington D, Cole P, Cooper TE, et al. Hydromorphone for neuropathic pain in adults. *Cochrane Database of Systematic Reviews* 2016, Issue 5. Art. No: CD011604. [DOI: 10.1002/14651858.CD011604.pub2]

39. Sommer C, Klose P, Welsch P, Petzke F, Häuser W. Opioids for chronic non-cancer neuropathic pain. An updated systematic review and meta-analysis of efficacy, tolerability and safety in randomized placebo-controlled studies of at least 4 weeks duration. *European Journal of Pain (London, England)* 2020;**24**:3-18. [DOI: 10.1002/ejp.1494]

40. Moore RA, Straube S, Aldington D. Pain measures and cut-offs - 'no worse than mild pain' as a simple, universal outcome. *Anesthesia* 2013;**68**:400-12.

41. de Souza MR, Henriques AT, Limberger RP. Medical cannabis regulation: an overview of models around the world with emphasis on the Brazilian scenario. *Journal of Cannabis Research* 2022;**4**:33. [DOI: 10.1186/s42238-022-00142-z]

42. Volkow ND, Baler RD, Compton WM, Weiss SR. Adverse health effects of marijuana use. *New England Journal of Medicine* 2014;**370**:2219-27.

43. Owens B. Drug development: the treasure chest. *Nature* 2015;**525**:S6-S8.

44. Hillard CJ, Weinlander KM, Stuhr KL. Contributions of endocannabinoid signaling to psychiatric disorders in humans: genetic and biochemical evidence. *Neuroscience* 2012;**204**:207-29.

45. Kalant H. Medicinal use of cannabis: history and current status. *Pain Research & Management* 2001;**6**:80-91.

46. Arthur P, Kalvala AK, Surapaneni SK, Singh MS. Applications of cannabinoids in neuropathic pain: an updated review. *Critical Reviews in Therapeutic Drug Carrier Systems* 2024;**41**:1-33. [DOI: 10.1615/CritRevTherDrugCarrierSyst.2022038592]

47. Alves P, Amaral C, Teixeira N, Correia-da-Silva G. Cannabis sativa: much more beyond Δ^9-tetrahydrocannabinol. *Pharmacology Research* 2020;**157**:104822. [DOI: 10.1016/j.phrs.2020.104822]

48. Guindon J, Hohmann AG. The endocannabinoid system and pain. *CNS & Neurological Disorders Drug Targets* 2009;**8**:403-29.

49. Lee MC, Ploner M, Wiech K, Bingel U, Wanigasekera V, Brooks J, et al. Amygdala activity contributes to the dissociative effect of cannabis on pain perception. *Pain* 2013;**134**:123-34.

50. Zhang J, Echeverry S, Lim TK, Lee SH, Shi XQ, Huang H. Can modulating inflammatory response be a good strategy to treat neuropathic pain? *Current Pharmaceutical Design* 2015;**21**:831-9.

51. Häuser W, Finn DP, Kalso E, Krcevski-Skvarc N, Kress HG, Morlion B, et al. European Pain Federation (EFIC) position paper on appropriate use of cannabis-based medicines and medical cannabis for chronic pain management. *European Journal of Pain (London, England)* 2018;**22**:1547-64. [DOI: 10.1002/ejp.1297]

52. Soliman N, Hohmann AG, Haroutounian S, Wever K, Rice AS, Finn D. Protocol for the systematic review and meta-analysis of studies in which cannabinoids were tested for antinociceptive effects in animal models of pathological or injury-related persistent pain. *Pain Reports* 2019;**4**:e766. [DOI: 10.1097/PR9.0000000000000076]

53. Krcevski-Skvarc N, Wells C, Häuser W. Availability and approval of cannabis-based medicines for chronic pain management and palliative/supportive care in Europe – a survey of the status in the chapters of the European Pain Federation. European Journal Pain 2018;**22**:440-54.

54. Häuser W, Petzke F, Fitzcharles MA. Efficacy, tolerability and safety of cannabis-based medicines for chronic pain management - an overview of systematic reviews. European Journal of Pain 2018;**22**:455-70.

55. Andreae MH, Carter GM, Shaparin N, Suslov K, Ellis RJ, Ware MA, et al. Inhaled cannabis for chronic neuropathic pain: a meta-analysis of individual patient data. *Journal of Pain* 2015;**16**(12):1221-32.

56. Finnerup NB, Attal N, Haroutounian S, McNicol E, Baron R, Dworkin RH, et al. Pharmacotherapy for neuropathic pain in adults: a systematic review and meta-analysis. *Lancet Neurology* 2015;**14**:162-73. [DOI: 10.1016/S1474-4422(14)70251-0]

57. Eisenberg E, Morlion B, Brill S, Häuser W. Medicinal cannabis for chronic pain: the Bermuda triangle of low-quality studies, countless meta-analyses and conflicting recommendations. *European Journal of Pain* 2022;**16**:1183-5. [DOI: 10.1002/ejp.1946]

58. IASP Presidential Task Force on cannabis and cannabinoid analgesia. International Association for the Study of Pain Presidential Task Force on cannabis and cannabinoid analgesia position statement. *Pain* 2021;**162**:S1-S2. [DOI: 10.1097/j.pain.0000000000002265]

59. Fisher E, Moore RA, Fogarty AE, Finn DP, Finnerup NB, Gilron I, et al. Cannabinoids, cannabis, and cannabis-based medicine for pain management: a systematic review of randomised controlled trials. *Pain* 2021;**162 (Suppl 1)**:S45-S66. [DOI: 10.1097/j.pain.0000000000001929]

60. Sainsbury B, Bloxham J, Pour MH, Padilla M, Enciso R. Efficacy of cannabis-based medications compared to placebo for the treatment of chronic neuropathic pain: a systematic review with meta-analysis. *Journal of Dental Anesthesia and Pain Medicine* 2021;**21**:479-506. [DOI: 10.17245/jdapm.2021.21.6.479]

61. Fitzcharles MA, Clauw DJ, Ste-Marie PA, Shir Y. The dilemma of medical marijuana use by rheumatology patients. *Arthritis Care & Research* 2014;**66**:797-801.

62. Weibel S, Popp M, Reis S, Skoetz N, Garner P, Sydenham E. Identifying and managing problematic trials: a research integrity assessment tool for randomized controlled trials in evidence synthesis. *Research Synthesis Methods* 2023;**14**:357-69. [DOI: 10.1002/jrsm.1599]

63. Dumville JC, Torgenson DJ, HewiTT DC. Reporting attrition in randomised controlled trials. *BMJ* 2006;**332**:967-71. [DOI: 10.1136/bmj.332.7547.969]

64. Deeks JJ, Higgins JP, Altman DG, McKenzie JE, Veroniki AA (editors). Chapter 10: Analysing data and undertaking meta-analyses [last updated November 2024]. In: Higgins JP, Thomas J,Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.5. Cochrane, 2024. Available from cochrane.org/handbook.

65. Cumpston M, Lasserson T, Flemyng E, Page MJ. Chapter III: Reporting the review [last updated August 2023]. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al (editors). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.5. Cochrane, 2024. Available from cochrane.org/handbook.

66. Higgins JP, Lasserson T, Thomas J, Flemyng E, Churchill R. Methodological Expectations of Cochrane Intervention Reviews. Cochrane: London, Version August 2023. Available from https://community.cochrane.org/mecir-manual.

67. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 2021;**372**:n71.

68. Truini A, Galeotti F, La Cesa S, Di Rezze S, Biasiotta A, et al. Mechanisms of pain in multiple sclerosis: a combined clinical and neurophysiological study. *Pain* 2012;**153**:2048-54. [DOI: 10.1016/j.pain.2012.05.024]

69. Koppel BS, Brust JC, Fife T, Bronstein J, Youssof S, Gronseth G, et al. Systematic review: efficacy and safety of medical marijuana in selected neurologic disorders: report of the Guideline Development Subcommittee of the American Academy of Neurology. *Neurology* 2014;**82**:1556-63.

70. Clauw D. What is the meaning of "small fiber neuropathy" in fibromyalgia? *Pain* 2015;**156**(11):2115-6.

71. Walitt B, Klose P, Fitzcharles MA, Häuser W. Cannabinoids for fibromyalgia. *Cochrane Database of Systematic Reviews* 2016, Issue 7. Art. No: CD011694. [DOI: 10.1002/14651858.CD011694]

72. Baron R, Binder A. How neuropathic is sciatica? The mixed pain concept [Wie neuropathisch ist die Lumboischialgie? Das gemischte Schmerzkonzept]. *Orthopäde* 2004;**33**:568-75.

73. McDonagh MS, Morasco BJ, Wagner J, Ahmed AY, Fu R, Kansagara D, et al. Cannabis-based products for chronic pain: a systematic review. *Annals of Internal Medicine* 2022;**175**:1143-53. [DOI: 10.7326/M21-4520]

74. Ye L, Cao Z, Wang W, Zhou N. New insights in cannabinoid receptor structure and signaling. *Current Molecular Pharmacology* 2019;**12**:239-48. [DOI: 10.21741874467212666190215112036]

75. Long JZ, Nomura DK, Vann RE, Walentiny DM, Booker L, Jin X, et al. Dual blockade of FAAH and MAGL identifies behavioral processes regulated by endocannabinoid crosstalk in vivo. *Proceedings of the National Academy of Science of the United States of America* 2009;**106**:20,270-5.

76. Cochrane Pain, Palliative and Supportive Care Group. PaPaS author and referee guidance. papas.cochrane.org/papas-documents (accessed 8 September 2025).

77. Dworkin RH, Turk DC, Wyrwich KW, Beaton D, Cleeland CS, Farrar JT, et al. Interpreting the clinical importance of treatment outcomes in chronic pain clinical trials: IMMPACT recommendations. *Journal of Pain* 2008;**9**(2):105-21. [DOI: 10.1016/j.jpain.2007.09.005]

78. International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH). Guideline for Good Clinical Practice E6(R2); November 2016. Available at https://database.ich.org/sites/default/files/E6_R2_Addendum.pdf (accessed prior to 04 December 2025).

79. International Council for Harmonisation. Medical Dictionary for Regulatory Activities Version 28.0. 2025. https://admin.new.meddra.org/sites/default/files/guidance/file/SMQ_intguide_28_0_English.pdf (accessed 8 September 2025).

80. Cochrane Central Register of Controlled Trials (CENTRAL). CENTRAL; 2025, Issue1 [CENTRAL; 2025, Issue1]. https://www.cochranelibrary.com/central (accessed 8 September 2025).

81. Moher D, Liberati A, Tetzlaff J, Altman DG; PRISMA Group (2009). Preferred reporting items for systematic reviews and meta-analyses: the PRISMA Statement. *PLoS Medicine* 2009;**6**(7):e1000097. [DOI: 10.1371/journal.pmed1000097]

82. ENTRUST-PE Network; O'Connell NE, Belton J, Crombez G, Eccleston C, Fisher E. Enhancing the trustworthiness of pain research: a call to action. *Journal of Pain* 2024;**16**:104763. [DOI: 10.1016/j.jpain.2024.104736]

83. Cochrane Library. Cochrane database of systematic reviews: editorial policies (last update 09 December 2024). www.cochranelibrary.com/cdsr/editorial-policies#problematicstudies (accessed 8 September 2025).

84. Birkinshaw H, Friedrich CM, Cole P, Eccleston C, Serfaty M, Stewart G, et al. Antidepressants for pain management in adults with chronic pain: a network meta-analysis. *Cochrane Database Systematic Reviews* 2023;**5**:CD014682. [DOI: 10.1002/14651858.CD014682.pub2]

85. Onakpoya IJ, Thomas ET, Lee JJ, Goldacre B, Heneghan CJ. Benefits and harms of pregabalin in the management of neuropathic pain: a rapid review and meta-analysis of randomised clinical trials. *BMJ Open* 2019;**9**:e023600. [DOI: 10.1136/bmjopen-2018-023600]

86. Soliman N, Moisset X, Ferraro MC, de Andrade DC, Baron R, Belton J, et al. NeuPSIG Review Update Study Group. Pharmacotherapy and non-invasive neuromodulation for neuropathic pain: a systematic review and meta-analysis. *Lancet Neurology* 2025;**24**:413-28. [DOI: 10.1016/S1474-4422(25)00068-7]

87. Berman JS, Symonds C, Birch R. Efficacy of two cannabis based medicinal extracts for relief of central neuropathic pain from brachial plexus avulsion: results of a randomised controlled trial. *Pain* 2004;**112**:299-306. [DOI: 10.1016/j.pain.2004.09.013]

88. Ellis RJ, Toperoff W, Vaida F, Van den Brande G, Gonzales J, Gouaux B, et al. Smoked medicinal cannabis for neuropathic pain in HIV: a randomized, crossover clinical trial. *Neuropsychopharmacology* 2009;**34**(3):672-80. [DOI: 10.1038/npp.2008.120;]

89. Frank B, Serpell MG, Hughes J, Matthews JN, Kapur D. Comparison of analgesic effects and patient tolerability of nabilone and dihydrocodeine for chronic neuropathic pain: randomised, crossover, double blind study. *BMJ (Clinical Research Ed.)* 2008;**336**:199-201. [DOI: 10.1136/bmj.39429.619653.80]

90. Langford RM, Mares J, Novotna A, Vachova M, Novakova I, Notcutt W, et al. A double-blind, randomized, placebo-controlled, parallel-group study of THC/CBD oromucosal spray in combination with the existing treatment regimen, in the relief of central neuropathic pain in patients with multiple sclerosis. *Journal of Neurology* 2013;**260**:984-97. [DOI: 10.1007/s00415-012-6739-4]

91. NCT00391079. Sativex versus placebo when added to existing treatment for central neuropathic pain in MS. https://clinicaltrials.gov/search?term=NCT00391079 (first posted 24 June 2013).

92. NCT00710424. A study of Sativex for pain relief due to diabetic neuropathy. clinicaltrials.gov/ct2/results?term=NCT00710424+&Search=Search (first posted 4 July 2008; last update posted 03 May 2023).

93. NCT01606176. A study to evaluate the effects of cannabis based medicine in patients with pain of neurological origin. clinicaltrials.gov/ct2/results?cond=&term=NCT01606176&cntry1=&state1=&recrs= (first posted 25 May 2012; last update posted 10 January 2023).

94. NCT01606202. A study of cannabis based medicine extracts and placebo in patients with pain due to spinal cord injury. Clinicaltrials.gov/Ct2/Results?term= NCT 01606202&Search=Search (first posted 25 May 2012; last update posted 10 January 2023).

95. Nurmikko TJ, Serpell MG, Hoggart B, Toomey PJ, Morlion BJ, Haines D. Sativex successfully treats neuropathic pain characterised by allodynia: a randomised, double-blind, placebo-controlled clinical trial. *Pain* 2007;**133**:210-20. [DOI: 10.1016/j.pain.2007.08.028]

96. NCT00711880. A study of sativex for relief of peripheral neuropathic pain associated with allodynia. https://clinicaltrials.gov/study/NCT00711880 (first posted 07 July 2008; last update posted 12 April 2023).

97. Rog DJ, Nurmikko TJ, Friede T, Young CA. Randomized, controlled trial of cannabis-based medicine in central pain in multiple sclerosis. *Neurology* 2005;**65**:812-9. [DOI: 10.1212/01.wnl.0000176753.45410.8b]

98. NCT01604265. A study of Sativex in the treatment of central neuropathic pain due to multiple sclerosis. https://clinicaltrials.gov/study/NCT01604265 (first posted 22 May 2012; last update posted 10 January 2023).

99. Schimrigk S, Marziniak M, Neubauer C, Kugler EM, Werner G, Abramov-Sommariva D. Dronabinol is a safe long-term treatment option for neuropathic pain patients. *European Neurology* 2017;**78**:320-9. [DOI: 10.1159/000481089]

100. Selvarajah D, Gandhi R, Emery CJ, Tesfaye S. Randomized placebo-controlled double-blind clinical trial of cannabis-based medicinal product (Sativex) in painful diabetic neuropathy:

depression is a major confounding factor. *Diabetes Care* 2010;**33**:128-30. [DOI: 10.2337/dc09-1029]

101. Serpell M, Ratcliffe S, Hovorka J, Schofield M, Taylor L, Lauder H, et al. A double-blind, randomized, placebo-controlled, parallel group study of THC/CBD spray in peripheral neuropathic pain treatment. *European Journal of Pain (London, England)* 2014;**18**:999-1012. [DOI: 10.1002/j.1532-2149.2013.00445.x]

102. NCT00710554. A study of Sativex for pain relief of peripheral neuropathic pain associated with allodynia. https://clinicaltrials.gov/study/NCT00710554 (first posted 03 July 2008; last update posted 03 May 2023).

103. Svendsen KB, Jensen TS, Bach FW. Does the cannabinoid dronabinol reduce central pain in multiple sclerosis? Randomised double blind placebo controlled crossover trial. *BMJ (Clinical Research Ed.)* 2004;**329**:253. [DOI: 10.1136/bmj.38149.566979.AE]

104. Toth C, Mawani S, Brady S, Chan C, Liu C, Mehina E, et al. An enriched-enrolment, randomized withdrawal, flexible-dose, double-blind, placebo-controlled, parallel assignment efficacy study of nabilone as adjuvant in the treatment of diabetic peripheral neuropathic pain. *Pain* 2012;**153**:2073-82. [DOI: 10.1016/j.pain.2012.06.024]

105. Xu DH, Cullen BD, Tang M, Fang Y. The effectiveness of topical cannabidiol oil in symptomatic relief of peripheral neuropathy of the lower extremities. *Current Pharmaceutical Biotechnology* 2020;**21**:390-402. [DOI: 10.2174/1389201020666191202111534]

106. Seevathee K, Kessomboon P, Manimmanakorn N, Luangphimai S, Thaneerat T, et al. Efficacy and safety of transdermal medical cannabis (THC: CBD: CBN formula) to treat painful diabetic peripheral neuropathy of lower extremities. *Medical Cannabis and Cannabinoids* 2025;**14**:1-14. [DOI: 10.1159/000542511]

107. D'Andre S, Novotny P, Walters C, Lewis-Peters S, Thomé S, Tofthagen CS. Topical cannabidiol for established chemotherapy-induced neuropathy: a pilot randomized placebo-controlled trial. *Cannabis and Cannabinoid Research* 2024;**9**:e1556-e1564. [DOI: 10.1089/can.2023.0253]

108. Eibach L, Scheffel S, Cardebring M, Lettau M, Özgür Celik M et al. Cannabidivarin for HIV-associated neuropathic pain: a randomized, blinded, controlled clinical trial. *Clinical Pharmacolology & Therapeutics* 2021;**109**:1055-62. [DOI: 10.1002/cpt.2016]

109. Hansen JS, Gustavsen S, Roshanisefat H, Kant M, Biering-Sørensen F, et al. Cannabis-based medicine for neuropathic pain and spasticity: a multicenter, randomized, double-blinded, placebo-controlled trial. *Pharmaceuticals (Basel, Switzerland)* 2023;**16**:1079. [DOI: 10.3390/ph16081079]

110. Lynch ME, Cesar-Rittenberg P, Hohmann AG. A double-blind, placebo-controlled, crossover pilot trial with extension using an oral mucosal cannabinoid extract for treatment of chemotherapy-induced neuropathic pain. *Journal of Pain and Symptom Management* 2014;**47**:166-73. [DOI: 10.1016/j.jpainsymman.2013.02.018]

111. Zubcevic K, Petersen M, Bach FW, Heinesen A, Enggaard TP, Almdal TP, et al. Oral capsules of tetra-hydro-cannabinol (THC), cannabidiol (CBD) and their combination in peripheral neuropathic pain treatment. *European Journal of Pain* 2023;**27**:492-506. [DOI: 10.1002/ejp.2072]

112. International Committee of Medical Journal Editors. Clinical trial registration in recommendations for the conduct, reporting, editing, and publication of scholarly work in medical journals; 2018. Available from http://www.icmje.org/icmje-recommendations.pdf (accessed 8 September 2025).

113. World Medical Association. World Medical Association Declaration of Helsinki: ethical principles for medical research involving human subjects. *Journal of the American Medical Association* 2013;**310**:2191-4. [DOI: 10.1001/jama.2013.281053]

114. European Union. Regulation (EU) 2017/745 of the European Parliament and of the Councilof 5 April 2017 on medical devices, amending Directive 2001/83/EC, Regulation (EC) No 178/2002 and Regulation (EC) No 1223/2009 and repealing Council Directives 90/385/EEC and 93/42/EEC. *Official Journal of the European Union* 2017;**L 117**:1-175.

115. EUPATI. Clinical trial approval in Europe. https://toolbox.eupati.eu/resources/clinical-trial-approval-in-europe (accessed 8 September 2025).

116. National Academies of Sciences, Engineering, and Medicine; Health and Medicine Division; Board on Population Health and Public Health Practice; Committee on the Health Effects of Marijuana. Chapter 15: Challenges and barriers in conducting cannabis research. In: The Health Effects of Cannabis and Cannabinoids: The Current State of Evidence and Recommendations for Research. Washington DC: National Academies Press, 2017. [DOI: 10.17226/24625]

117. Higgins JP, Savović J, Page MJ, Elbers RG, Sterne JA. Chapter 8: Assessing risk of bias in a randomized trial [last updated October 2019]. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al (editors). *Cochrane Handbook for Systematic Reviews of Interventions* version 6.5 (updated August 2024). Cochrane, 2024. Available from cochrane.org/handbook.

118. Häuser W, Welsch P, Radbruch L, Fisher E, Bell RF, Moore RA. Cannabis-based medicines and medical cannabis for adults with cancer pain. *Cochrane Database Systematic Reviews* 2023;**6**:CD014915. [DOI: 10.1002/14651858.CD014915.pub2]

119. Schaefert R, Welsch P, Klose P, Sommer C, Petzke F, Häuser W. Opioids in chronic osteoarthritis pain. A systematic review and meta-analysis of efficacy, tolerability and safety in randomized placebo-controlled studies of at least 4 weeks duration [Opioide bei chronischem Arthroseschmerz. Systematische Übersicht und Metaanalyse der Wirksamkeit, Verträglichkeit und Sicherheit in randomisierten, placebokontrollierten Studien über mindestens 4 Wochen]. *Schmerz (Berlin, Germany)* 2015;**29**:47-59.

120. McQuay H, Moore R. An Evidence-Based Resource for Pain Relief. Oxford: Oxford University Press, 1998. [ISBN: 0-19-263048-2]

121. Moore RA, Barden J, Derry S, McQuay HJ. Managing potential publication bias. In: McQuay HJ, Kalso E, Moore RA, editor(s). Systematic Reviews in Pain Research: Methodology Refined. Seattle: IASP Press, 2008:15-24. [ISBN: 978-0-931092-69-5]

122. Cohen J. Statistical Power Analysis for the Behavioral Sciences. Hillsdale: Lawrence Erlbaum Associates, 1988.

123. Fayers PM, Hays RD. Don't middle your MIDs: regression to the mean shrinks estimates of minimally important differences. *Quality of Life Research* 2014;**23**:1-4.

124. Furukawa TA, Cipriani A, Barbui C, Brambilla P, Watanabe N. Imputing response rates from means and standard deviations in meta-analyses. *International Clinical Psychopharmacology* 2005;**20**:49-52.

125. Review Manager (RevMan). Version 9.16.1. The Cochrane Collaboration, 2025. Available from revman.cochrane.org.

126. L'Abbé KA, Detsky AS, O'Rourke K. Meta-analysis in clinical research. *Annals of Internal Medicine* 1987;**107**:224-33.

127. Higgins JP, Thompson SG, Deeks JJ, Altman DG. Measuring inconsistency in meta-analyses. *BMJ (Clinical Research Ed.)* 2003;**327**:557-60.

128. Schünemann HJ, Higgins JP, Vist GE, Glasziou P, Akl LA, Skoetz N, et al. Chapter 14: Completing 'Summary of findings' tables and grading the certainty of the evidence [last updated August 2023]. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al (editors). *Cochrane Handbook for Systematic Reviews of Interventions* version 6.5. Cochrane, 2024. Available from cochrane.org/handbook.

129. Ware MA, Wang T, Shapiro S, Robinson A, Ducruet T, Huynh T et al. Smoked cannabis for chronic neuropathic pain: a randomized controlled trial. *Canadian Medical Association Journal* 2010;**182**:E694-701. [DOI: 10.1503/cmaj.091414]

130. Abrams DI, Jay CA, Shade SB, Vizoso H, Reda H, Press S, et al. Cannabis in painful HIV-associated sensory neuropathy: a randomized placebo-controlled trial. *Neurology* 2007;**68**:515-21.

131. Corey-Bloom J, Wolfson T, Gamst A, Jin S, Marcotte TD, Bentley H, et al. Smoked cannabis for spasticity in multiple sclerosis: a randomized, placebo-controlled trial. *Canadian Medical Association Journal* 2012;**184**:1143-50.

132. Karst M, Salim K, Burstein S, Conrad I, Hoy L, Schneider U. Analgesic effect of the synthetic cannabinoid CT-3 on chronic neuropathic pain: a randomized controlled trial. *JAMA* 2003;**290**:1757-62.

133. Notcutt W, Price M, Miller R, Newport S, Phillips C, Simmons S, et al. Initial experiences with medicinal extracts of cannabis for chronic pain: results from 34 'N of 1' studies. *Anaesthesia* 2004;**59**:440-52.

134. Novotna A, Mares J, Ratcliffe S, Novakova I, Vachova M, Zapletalova O, et al. Arandomized, double-blind, placebo-controlled, parallel-group, enriched-design study of nabiximols* (Sativex(®)), as add-on therapy, in subjects with refractory spasticity caused by multiple sclerosis. *European Journal of Neurology* 2011;**18**:1122-31.

135. Rintala DH, Fiess RN, Tan G, Holmes SA, Bruel BM. Effect of dronabinol on central neuropathic pain after spinal cord injury: a pilot study. *American Journal of Physical Medicine & Rehabilitation* 2010;**89**:840-8.

136. Turcotte D, Doupe M, Torabi M, Gomori A, Ethans K, Esfahani F, et al. Nabilone as an adjunctive to gabapentin for multiple sclerosis-induced neuropathic pain: a randomized controlled trial. *Pain Medicine (Malden, Mass.)* 2015;**16**:149-59.

137. Wade DT, Robson P, House H, Makela P, Aram J. A preliminary controlled study to determine whether whole-plant cannabis extracts can improve intractable neurogenic symptoms. *Clinical Rehabilitation* 2003;**17**:21-9.

138. Wade DT, Makela P, Robson P, House H, Bateman C. Do cannabis-based medicinal extracts have general or specific effects on symptoms in multiple sclerosis? A double-blind, randomized, placebo-controlled study on 160 patients. *Multiple Sclerosis (Houndmills, Basingstoke, England)* 2004;**10**:434-41.

139. Wallace MS, Marcotte TD, Umlauf A, Gouaux B, Atkinson JH. Efficacy of inhaled cannabis on painful diabetic neuropathy. *Journal of Pain* 2015;**16**:616-27.

140. Wilsey B, Marcotte T, Tsodikov A, Millman J, Bentley H, Donaghe H. A randomized, placebo-controlled, crossover trial of cannabis cigarettes in neuropathic pain. *Journal of Pain* 2008;**9**:506-21.

141. Wilsey B, Marcotte T, Deutsch R, Gouaux B, Sakai S, Gouaux B, et al. Low-dose vaporized cannabis significantly improves neuropathic pain. *Journal of Pain* 2013;**14**:136-48.

142. Wissel J, Haydn T, Müller J, Brenneis C, Berger T, Berger T. Low dose treatment with the synthetic cannabinoid Nabilone significantly reduces spasticity-related pain: a double-blind placebo-controlled cross-over trial. *Journal of Neurology* 2006;**253**:1337-41.

143. Zajicek J, Fox P, Sanders H, Wright D, Vickery J, Nunn A, et al. Cannabinoids for treatment of spasticity and other symptoms related to multiple sclerosis (CAMS study): multicentre randomized placebo-controlled trial. *Lancet* 2003;**362**:1517-26.

144. Zajicek JP, Hobart JC, Slade A, Barnes D, Mattison PG; MUSEC Research Group. Multiple sclerosis and extract of cannabis: results of the MUSEC trial. *Journal of Neurology, Neurosurgery and Psychiatry* 2012;**83**:1125-32.

145. Collin C, Ehler E, Waberzinek G, Alsindi Z, Davies P et al. A double-blind, randomized, placebo-controlled, parallel-group study of Sativex, in subjects with symptoms of spasticity due to multiple sclerosis. *Neurology Research* 2010;**32**:451-9. [DOI: 10.1179/016164109X12590518685660]

146. Leocani L, Nuara A, Houdayer E, Schiavetti I, Del Carro U. Sativex and clinical–neurophysiological measures of spasticityin progressive multiple sclerosis. *Journal of Neurology* 2015;**262**:2520-7. [DOI: 10.1007/s00415-015-7878-1]

147. Markovà J, Essner U, Akmaz B, Marinelli M, Trompke C et al. Sativex® as add-on therapy vs. further optimized first-line ANTispastics (SAVANT) in resistant multiple sclerosis spasticity: a double-blind, placebo-controlled randomised clinical trial. *International Journal of Neuroscience* 2019;**129**:119-28. [DOI: 10.1080/00207454.2018.1481066]

148. Kittithamvongs P, Anantasinkul P, Siripoonyothai S, Anantavorasakul N, Malungpaishrope K, Uerpairojkit C et al. Does Cannabis-based Medicine Improve Pain and Sleep Quality in Patients With Traumatic Brachial Plexus Injuries? A Triple-blind, Crossover, Randomized Controlled Trial. *Clinical Orthopaedics and Related Research* 2025;**483**:228-34. [DOI: 10.1097/CORR.0000000000003221]

149. NCT01037088. Effects of vaporized marijuana on neuropathic pain. https://clinicaltrials.gov/study/NCT01037088 Last update posted 2018-01-31.

150. NCT03099005. Effect of cannabis and endocannabinoids on HIV neuropathic pain. https://clinicaltrials.gov/study/NCT03099005 First and last update posted 2024-06-18.

151. Weizman L, Sharon H, Dayan L, Espaniol J, Brill S, Nahman-Averbuch H et al. Oral Delta-9-Tetrahydrocannabinol (THC) Increases Parasympathetic Activity and Supraspinal Conditioned Pain Modulation in Chronic Neuropathic Pain Male Patients: a Crossover, Double-Blind, Placebo-Controlled Trial. *CNS Drugs* 2024;**38**:375-85. [DOI: 10.1007/s40263-024-01085-0]

152. NCT00699634. Nabilone for the treatment of phantom limb pain. clinicaltrials.gov/ct2/show/NCT00699634?term=NCT00699634&rank=1 (first posted 18 June 18).

153. NCT01035281. Efficacy study of nabilone in the treatment of diabetic peripheral neuropathic pain. clinicaltrials.gov/ct2/show/NCT01035281?term=NCT01035281&rank=1 (first posted 18 December 2009).

154. NCT01222468. Effect of cannabinoids on spasticity and neuropathic pain in spinal cord injured persons. clinicaltrials.gov/ct2/show/NCT01222468?term=NCT01222468&rank=1 (first posted 18 October 2010).

155. NCT02460692. Trial of dronabinol and vaporized cannabis in chronic low back pain [Trial of Dronabinol and Vaporized Cannabis in Chronic Low Back Pain]. https://clinicaltrials.gov/study/NCT02460692 2015;**Last update posted 2023-09-28**.

156. van Amerongen G, Kanhai K, Baakman AC, Heuberger J, Klaassen E et al. Effects on Spasticity and Neuropathic Pain of an Oral Formulation of Δ9-tetrahydrocannabinol in Patients WithProgressive Multiple Sclerosis. *Clinical Therapeutics* 2018;**40**:1467-82. [DOI: 10.1016/j.clinthera.2017.01.016]

157. Dechartres A, Trinquart L, Boutron I, Ravaud P. Influence of trial sample size on treatment effect estimates: meta-epidemiological study. *BMJ (Clinical Research Ed.)* 2013;**346**:f2304. [DOI: 10.1136/bmj.f2304]

158. Dechartres A, Altman DG, Trinquart L, Boutron I, Ravaud P. Association between analytic strategy and estimates of treatment outcomes in meta-analyses. *JAMA* 2014;**312**:623-30. [DOI: 10.1001/jama.2014.8166]

159. Moore RA, Gavaghan D, Tramèr MR, Collins SL, McQuay HJ. Size is everything - large amounts of information are needed to overcome random effects in estimating direction and magnitude of treatment effects. *Pain* 1998;**78**(3):209-16. [DOI: 10.1016/S0304-3959(98)00140-7]

160. Nüesch E, Trelle S, Reichenbach S, Rutjes AW, Tschannen B, Altman DG, et al. Small study effects in meta-analyses of osteoarthritis trials: meta-epidemiological study. *BMJ (Clinical Research Ed.)* 2010;**341**:c3515. [DOI: 10.1136/bmj.c3515]

161. Thorlund K, Imberger G, Walsh M, Chu R, Gluud C, Wetterslev J, et al. The number of patients and events required to limit the risk of overestimation of intervention effects in meta-analysis: a simulation study. *PLoS ONE* 2011;**6**(10):e25491. [DOI: 10.1371/journal.pone.0025491]

162. AlBalawi Z, McAlister FA, Thorlund K, Wong M, Wetterslev J. Random error in cardiovascular meta-analyses: how common are false positive and false negative results? *International Journal of Cardiology* 2013;**168**(2):1102-7. [DOI: 10.1016/j.ijcard.2012.11.048]

163. Turner RM, Bird SM, Higgins JP. The impact of study size on meta-analyses: examination of underpowered studies in Cochrane Reviews. *PLoS ONE* 2013;**8**(3):e59202. [DOI: 10.1371/journal.pone.0059202]

164. Elbourne DR, Altman DG, Higgins JP, Curtin F, Worthington HV, Vail A. Meta-analyses involving cross-over trials: methodological issues. *International Journal of Epidemiology* 2002;**31**(1):140-9.

165. EMA. Guideline on the clinical development of medicinal products intended for the treatment of pain (15 December 2016). https://www.ema.europa.eu/en/documents/scientific-guideline/guideline-clinical-development-medicinal-products-intended-treatment-pain-first-version_en.pdf (accessed 2 May 2025).

166. Ware MA, Wang T, Shapiro S, Collet JP; COMPASS study team. Cannabis for the management of pain: assessment of safety study (COMPASS). *Journal of Pain* 2015;**16**:1233-42.

167. Baron R, Maier C, Attal N, Binder A, Bouhassira D, Cruccu G, et al. Peripheral neuropathic pain: a mechanism-related organizing principle based on sensory profiles. *Pain* 2017;**158**:261-72.

168. Tyson SF, Brown P. How to measure pain in neurological conditions? A systematic review of psychometric properties and clinical utility of measurement tools. *Clinical Rehabilitation* 2014;**28**:669-86.

169. Mintzes B, Lexchin J, Quintano AS. Clinical trial transparency: many gains but access to evidence for new medicines remains imperfect. *British Medical Bulletin* 2015;**116**:43-53.

170. Andersohn F, Garbe E. Pharmacoepidemiological research with large health databases. *Bundesgesundheitsblatt, Gesundheitsforschung, Gesundheitsschutz* 2008;**51**:1134-55.

171. Moore A, Straube S, Fisher E, Eccleston C. Cannabidiol (CBD) products for pain: ineffective, expensive, and with potential harms. *Journal of Pain* 2024;**25**:833-42. [DOI: 10.1016/j.jpain.2023.10.009]

172. Oliveira RA, Baptista AF, Sá KN, Barbosa LM, Nascimento O, Listik C, et al. Pharmacological treatment of central neuropathic pain: consensus of the Brazilian Academy of Neurology [Tratamento farmacológico da dor neuropática central: consenso da AcademiaBrasileira de Neurologia]. *Arquivos de Neuro-Psiquiatria* 2020;**14**:741-52. [DOI: 10.1590/0004-282X20200166]

173. Geißler J, Isham E, Hickey G, Ballard C, Corbett A, Lubbert C. Patient involvement in clinical trials. *Communications Medicine (London)* 2022;**2**:94. [DOI: 10.1038/s43856-022-00156-x]

174. Häuser W, Bartram C, Bartram-Wunn E, Tölle T. Adverse events attributable to nocebo in randomized controlled drug trials in fibromyalgia syndrome and painful diabetic peripheral neuropathy: systematic review. *Clinical Journal of Pain* 2012;**28**:437-51.

175. Cepeda MS, Berlin JA, Gao CY, Wiegand F, Wada DR. Placebo response changes depending on the neuropathic pain syndrome: results of a systematic review and meta-analysis. *Pain Medicine (Malden, Mass.)* 2012;**13**:575-95. [DOI: 10.1111/j.1526-4637.2012.01340.x]

176. Finnerup NB, Haroutounian S, Kamerman P, Baron R, Bennett DL, Bouhassira D, et al. Neuropathic pain: an updated grading system for research and clinical practice. *Pain* 2016;**157**:1599-606. [DOI: 10.1097/j.pain.0000000000000492]

177. Mücke M, Phillips T, Radbruch L, Petzke F, Häuser W. Cannabinoids for chronic neuropathic pain. *Cochrane Database of Systematic Reviews* 2016, Issue 5. Art. No: CD012182. [DOI: 10.1002/14651858.CD012182]

ADDITIONAL TABLES

Table 1. Overview of included studies and syntheses

Study ID	Study design; duration of randomised period	Setting	Reference medication/comparator	Type of neuropathic pain	Number of participants randomised/Attrition rate	Population: mean age; % female	Outcomes with data available for syntheses*
Berman 2004	Cross-over; 10–14 days each period	Single centre, UK	THC only and THC/CBD balanced oromucosal/placebo	Peripheral (plexus root injury)	48/6%	39;4	1,3,4,5,6,7,8,9,10
D'Andre 2024	Cross-over; 2 weeks each period	Single centre, USA	CBD only topical/placebo	Peripheral (chemotherapy-induced PNP)	40/14%	63;63	1,2,3,4,5,6,7,10,11,12,13
Eibach 2021	Cross-over; 4 weeks each period	Single centre, Germany	CBDV only orally/ placebo	Peripheral (HIV-associated PNP)	34/11%	52;0	1,3,4,5,6,7,10,11,12,13
Ellis 2009	Cross-over; 2 weeks each period	Single centre; USA	THC (probably) only inhaled/ placebo	Peripheral (HIV-associated PNP)	34 /18%	49;0	3,5,6
Frank 2008	Cross-over; 6 weeks each period	Multicentre, UK	THC oral/ dihydrocodeine	Peripheral (different aetiologies)	96/33%	51;49	1,3,4,5,6,11 (qualitative synthesis)
Hansen 2023	Parallel, 6 weeks	Multicentre, Denmark	THC only, THC/CBD balanced and CBD only, all orally/ placebo	Mixed (central (MS) and peripheral (spinal cord injury))	134/20%	51 to 52; 65 to 90	1 to 13
Langford 2013	Parallel 14 weeks; withdrawal 4 weeks	Multicentre, different European countries, Canada	THC/CBD balanced oromucosal/placebo	Central (MS)	339/12%	48;68	1 to 13
Lynch 2014	Cross-over; 4 weeks each period	Single centre, Canada	THC/CBD balanced oromucosal/placebo	Peripheral (chemotherapy-induced PNP)	18/11%	56;83	1,3,4,5,6,7,9,12,13
NCT00710424	Parallel; 14 weeks	Multicentre, different European countries	THC/CBD balanced oromucosal/placebo	Peripheral (pDPN)	297/ 23%	61;38	2,3,4,5,6,7,8,10,12,13
NCT01606176	Parallel; 3 weeks	Multicentre, UK	THC/CBD balanced oromucosal/placebo	Mixed (central (MS) and peripheral (different aetiologies))	70/10%	52;56	2,3,4,6,7,8,11,12,13

Table 1. Overview of included studies and syntheses *(Continued)*

NCT01606202	Parallel; 3 weeks	Multicentre, Romania	THC/CBD balanced oromucosal/placebo	Central (spinal cord injury)	116/9%	49;23	3,4,6,11,12,13
Nurmikko 2007	Parallel; 5 weeks	Multicentre; Belgium, UK	THC/CBD balanced oromucosal/placebo	Peripheral (different aetiologies)	125 /16%	52;56	1 to 13
Rog 2005	Parallel; 5 weeks	Single centre; UK	THC/CBD balanced oromucosal/placebo	Central (MS)	66/3%	48;77	1,2,3,4,5,6,7,9,11,12,13
Schimrigk 2017	Parallel; 10 weeks	Multicentre; Austria, Germany	THC only orally/placebo	Central (MS)	240/30%	48;71	1,3,4,5,6,11,12,13
Seevathee 2025	Parallel; 12 weeks	Single centre; Thailand	THC-dominant, low-dose CBD and cannabinol, transdermal/placebo	Peripheral (pDPN)	100/7%	63;72	1,5,6,11
Sevalrajah 2010	Parallel; 10 weeks	Single centre; UK	THC/CBD balanced oromucosal/placebo	Peripheral (pDPN)	30/23%	58;27	1,5,6,7,10
Serpell 2014	Parallel; 15 weeks	Multicentre; Canada, different European countries	THC/CBD balanced oromucosal/placebo	Peripheral (different aetiologies)	246/30%	58;66	2 to 13
Svendsen 2004	Cross-over; 3 weeks each period	Single centre; Denmark	THC only orally/placebo	Central (multiple sclerosis)	25/0%	50;58	1,3,4,5,6,7,8,9,10,11,12,13
Toth 2021	Withdrawal; 5 weeks	Single centre; Canada	THC only orally/ placebo	Peripheral (pDPN)	26/4%	61;62	1,2,3,4,6,7,8,9,10,11 (qualitative synthesis)
Xu 2020	Cross-over; 4 weeks each period	Single centre; USA	CBD only topical/placebo	Peripheral (pDPN)	29/21%	70;27	1,4,5,6,11,12,13
Zubcevic 2023	Parallel; 10 weeks	Multicentre; Denmark	THC only, THC/CBD balanced,CBD only orally/placebo	Peripheral (different aetiologies)	115/19%	62 to 68; 46 to 70	1,3,4,5,6,10,11,12,13

CBD: cannabidiol; **CBDV:** cannabidivarin: **HIV:** human immunodeficiency virus; **MS:** multiple sclerosis; **pDPN:** painful diabetic polyneuropathy; **PNP:** polyneuropathy; **THC:** tetrahydrocannabinol

***Outcomes key**

1: Pain relief of 50% or greater

2: Patient Global Impression of Change rating of 'much' or 'very much' improved

3: Withdrawals due to adverse events

4: Frequency of serious adverse events
5: Pain relief of 30% or greater
6: Mean pain intensity
7: Health-related quality of life
8: Sleep problems
9: Psychological distress
10: Withdrawals due to lack of efficacy
11: Frequency of any adverse events
12: Frequency of nervous system-related adverse events
13: Frequency of psychiatric disorder-related events